

Model Logic: Design Choices, Justifications and Alternatives

Cancer Screening Microsimulation Model (MedicalModel2024)

Model documentation

October 10, 2026

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1 Purpose, Scope and How to Read This Document

This document describes *what the model does and why it is built that way*. It is deliberately neither a code manual nor a port review. For almost every element of the model there is more than one defensible way to do it, so the useful information is not only the choice that was made but the alternatives that were rejected and the reason for rejecting them — because those reasons determine where the model is strong, and where a different dataset or a different question would force a different design.

Every modelling element is presented with the same three-part template:

1. **Mechanism** — the rule the model actually applies, stated so that it could be re-implemented from this description alone.
2. **Why this design** — what the choice buys: identifiability from the available data, epidemiological plausibility, interpretability, or computational cost.
3. **Alternatives** — other defensible designs, why each was not used here, and what would have to change to use it.

Two sibling documents cover the other angles and are not repeated here:

- `model_for_dummies.pdf` — a plain-language walkthrough of the model and the web interface.
- `python_port_analysis.pdf` — the technical and statistical review: state variables, statistical corrections, performance, and methodological improvements.

Reference configuration. Every number quoted below as an illustration comes from one configuration: the GLOBOCAN colorectal datasets (USA aggregate and staging files), $N = 20\,000$ agents, seed 2022, 30 simulated years, screening at ages 50–60 every second year with 50% participation. That run yields 1 142 cancer cases, 200 patients cured clinically, 5 cured after screen detection, 12 334 survivors at the end, and a total cost of 13 509 units. Numbers from other runs are not comparable one-to-one, because the model is stochastic.

A note on honesty. Where an element exists in the code but has no effect on the results, it is listed explicitly in Section 13 rather than silently omitted. Where the model *cannot* determine a quantity from the available data, that is stated at the point where it matters.

2 Model Class, Time and Population Frame

2.1 Why an agent-based microsimulation

Mechanism. The model creates N independent agents. Each agent is given one complete cancer history — whether and when a first malignant cell appears, how fast the tumour grows, what stage it has reached by the time of clinical diagnosis, whether treatment cures it, and at what age it dies — and is then followed year by year. Nothing is aggregated into fractions; the population statistics are obtained by counting agents after the fact.

Why this design. Four properties of the question force this design.

(i) *The answer is a difference of tails, not a mean.* Screening is justified by a small number of lives shifted out of the cancer-death column, and by the years of life those

people gain. Both are properties of individual counterfactuals, and they are exactly the quantities a model of averages cannot represent: the same individual must be observable with and without the intervention. The model therefore stores *both* death ages for every patient (untreated and screening-affected), so the survivors-per-year curve is computed from within-individual differences rather than from two independent populations. Heterogeneity and the tail go together: the people who benefit are not the average person.

(ii) *Screening is history-dependent.* Who is screened at age 58 depends on participation at 56, on whether a cancer was already found, and on the age window. In a state-based model this forces the state space to carry the whole screening history. In a microsimulation it is a few boolean flags per agent.

(iii) *The heterogeneity is structural, not decorative.* Individual growth rate (B_{ind}), per-stage aggressiveness, age-dependent cure efficiency and per-stage treatment efficiency all act multiplicatively on the outcome. Representing any of them faithfully in a cohort model means multiplying the number of states by the number of levels; combining them is combinatorially explosive.

(iv) *The model must reproduce a stage mix, not a mean stage.* The stage at diagnosis is a consequence of how long the tumour has been growing and how fast, so it is inherently an individual-level crossing event (Section 6.3).

Alternatives.

- **Cohort (deterministic) Markov / multi-state life table.** The classical alternative, used for many cancer screening analyses because it is fast, deterministic and easy to communicate. Rejected because it cannot carry the per-individual counterfactual that the “lives saved” output is defined by, and because the state space grows multiplicatively with heterogeneity.
- **Age-structured PDE (McKendrick–von Foerster).** Analytically tractable and fast for mean-field questions. Rejected for the same reason, plus a practical one: screening makes the transfer rates time- and history-dependent, which turns the PDE into a large system.
- **Renewal equation / integral projection model.** Very efficient for incidence and stage shift. Rejected because cause-specific mortality and the within-individual trade of cancer death against other-cause death are awkward to express.
- **Semi-Markov with sojourn-time distributions.** A good compromise and a genuine option. Rejected here because the stage-crossing mechanism is already a semi-Markov process with an explicit growth model; adding a second latent progression structure would double-count.
- **Compartmental ODE / system dynamics.** Rejected: no individual variability, no screening history, and “who benefits” becomes unanswerable.
- **Decision tree or Bayesian network.** Appropriate for a single one-off decision, not for a lifetime intervention with competing risks.

2.2 Why one-year discrete time steps

Mechanism. Time is the integer year counter `current_date`. Ages are integers (age = `current_date - date_birth`). At each step the model (a) removes the agents whose death age equals their current age, (b) runs a screening round if the year is on the screening grid, and (c) updates the statistics. All events within a year are resolved at year boundaries.

Why this design. The time step is chosen to match the granularity of the calibration data and of the intervention being modelled, not for convenience. The aggregate data are annual age-banded tables; the programme being evaluated is defined on an annual grid (“invite at ages 50–60 every two years”); and the natural-history processes involved are slow compared with a year, with one exception noted below. Annual integer ages make three internal devices exact rather than approximate: the death comparison is an integer equality, the stage at screen detection is a lookup of the stored crossing age against the current age, and the exported agent table is directly readable.

Why this design. The price is explicit: within a year, events are simultaneous and unordered, no event can occur twice in the same year, and a process faster than a year — the terminal phase after the last stage crossing — is discretised. That last point is not a footnote here: the fitted cancer-death hazard gives a mean terminal survival of roughly three years (Section 14), so a one-year grid is adequate but not generous.

Alternatives.

- **Continuous time with exponential clocks (Gillespie-style next-event loop).** Exact within-year dynamics and exact screening instants; requires an exponential assumption for every transition and a much larger implementation surface. This is the most defensible upgrade if the timing of the terminal phase ever becomes the object of study.
- **Monthly or quarterly steps.** Finer, but multiplies cost by 12 or 4 while the calibration data cannot support sub-annual targets, so the extra resolution would not be controlled by data.
- **Event-driven simulation with a priority queue.** Natural for continuous time; more machinery than the annual grid needs.
- **Hybrid: annual screening grid with continuous natural history.** A reasonable middle ground, but it complicates reproducibility and the exported agent table for no gain at the current level of data support.

2.3 Why a closed cohort and a 30-year horizon

Mechanism. There are no births and no migration: the population ages and dies. The initial age structure is taken from the observed population, so the cohort already contains all ages at $t = 0$. The horizon is `years_to_simulate` (30 by default).

Why this design. A closed cohort isolates the effect being estimated. If births and migration were simulated, the observed mortality difference between the screening and no-screening scenarios would mix the screening effect with demographic drift, and the model would need fertility and migration assumptions it cannot validate against anything. Because the initial age distribution is cross-sectional — people enter at their current age — the cohort is already realistic at $t = 0$, so no burn-in is needed and the first year of output is usable; this matters for a screening window that opens at age 50. Thirty years covers the whole screening window twice over and allows the mortality difference to accumulate, while keeping the run interactive: the reference 20 000-agent, 30-year run completes in a couple of seconds.

Alternatives.

- **Birth cohort followed from age 0.** Cleanest for lifetime outcomes and for interactions with childhood exposures; requires a burn-in of a century and many more agents to reach the same statistical power in the ages 50–80 where the decisions are.
- **Open population with demographic projection.** More realistic for long-run forecasts; introduces fertility and migration assumptions this model has no data to support, and contaminates the screening contrast.
- **Steady-state (stationary) population.** Attractive if only long-run rates matter; adds a demographic-equilibrium assumption without helping the screening question.
- **Hybrid entry at mid-life.** Entering a cohort at age 40–50 would shorten runs further, but discards the interaction between the age at which screening starts and the pre-screening history.

3 Demography: Initial Age Structure and Competing Mortality

3.1 Initial age structure

Mechanism. Every agent is assigned a starting age by sampling from the observed population age structure: the aggregate data file’s **population** column, one value per year of age, is normalised and used as a discrete distribution; ages are drawn by the inverse-CDF method (a uniform draw is located in the cumulative array). The agent’s birth year is then set to minus that age, so absolute age is simply “time since birth”.

Why this design. The cross-sectional age structure is the correct initial condition for a closed cohort: it fixes both the number of people at risk and the age profile of that risk at the moment the simulation starts. Using the file as a probability mass rather than as absolute counts means the file’s units cancel, so the model does not depend on whether the column is in persons, thousands or percentages. Single years of age are used because the data are already available at that resolution; the model imposes no binning of its own, so any loss of detail is the data’s choice rather than the model’s.

Alternatives.

- **A uniform or single-peaked age distribution.** Rejected: the age distribution is the dominant driver of both incidence and mortality, so an unrealistic one would corrupt every output.
- **Age bands with a step function.** Cheaper and closer to how source data are often published; it loses intra-band resolution, which matters near the screening window.
- **A fitted parametric age profile (Gompertz, Weibull, or a spline).** Smooths sampling noise and extrapolates, but imposes a shape the data do not require, and any misfit propagates into every rate.
- **Empirical microdata (a census or survey sample of individual ages).** The most faithful option; it needs data the packaged model does not ship with, and the aggregate route already reproduces the marginal distribution.

- **Starting everyone at the same age (e.g. all at 50).** Rejected: it destroys the age composition, removes the youngest and oldest, and makes the screening window a degenerate cohort boundary.

3.2 Non-cancer mortality as a competing risk

Mechanism. Each agent also receives a *natural death age*, sampled in the same way from a distribution built from the observed annual all-cause death counts (`deaths` all divided by the total population, normalised). Agents whose sampled death age is not greater than their starting age are resampled once; if the redraw still fails the death age is set to starting age +1, so the agent dies in the first simulated year. Thereafter the natural death age is fixed, and everything the model does is compared against it.

Why this design. Addressing non-cancer mortality as a single fixed individual “death age”, drawn once, buys three things.

(i) *A competing risk becomes an arithmetic comparison.* With one natural death age and one cancer death age per agent, everything that depends on their interaction — did the cancer or something else kill this person first, was this person cured, did screening save them, how many life-years were gained — is a comparison of two numbers. This is what makes the paired screening/no-screening accounting exact rather than approximate: there is no control population to re-simulate and no extra sampling noise in the contrast.

(ii) *It is distributionally equivalent to a survival model without the bookkeeping.* Drawing the death age from the life table is marginally the same as removing an annual fraction of survivors, but it replaces a running state variable and an iterative update with one stored integer.

(iii) *The conditional redraw is the right correction.* Because agents enter at their current age, their death age must be conditional on already being alive; resampling the minority whose first draw falls below the entry age implements exactly that truncation, and the fallback (age + 1) guarantees a well-defined one-year life instead of a zero or negative lifespan.

Why this design. *A consequence that must be stated plainly.* An agent is removed from the population only at its natural death age. A recorded cancer death does *not* remove the agent; cancer deaths are counted as cause-specific events in the statistics layer. The age structure therefore follows the all-cause life table exactly, and the at-risk denominator used for cause-specific rates still contains agents who already have a cancer death recorded. This is the standard device of estimating a cause-specific hazard with the general-population denominator, and it is correct under the usual assumption that the two causes act independently — that removing one cause does not change the force of the other. It would be the wrong device if a cancer death were assumed to change the subsequent risk of dying from other causes, and it is one of the assumptions a validation exercise should probe.

Alternatives.

- **An annual Bernoulli draw with a running survival flag.** Distributionally equivalent for a single cause, but it makes the death age an output rather than

an input, which makes the paired counterfactual much harder, and it adds a state variable that must be kept consistent.

- **A parametric mortality law (Gompertz–Makeham, Heligman–Pollard, Lee–Carter).** Smooth, compact, extrapolable; unnecessary here because the whole age range (0–110) is observed, so a fitted law could only add misfit.
- **Cause-deleted (other-cause) life tables.** Conceptually the right construction for a cancer model: remove cancer deaths from the background so the two causes are not double-counted. Not used because the packaged aggregate file supplies all-cause deaths only — the required input is a cause-specific death column by age. The helper that converts annual risks into a death-age distribution (`risks_to_distribution`, $P(T = a) = q(a) \prod_{j < a} (1 - q(j))$) is present in the code but never called: the model uses observed deaths directly.
- **Frailty models (a Gamma-distributed individual hazard multiplier).** Captures mortality selection — the frail die first, so survivors are a selected subgroup — and is a well-established correction. Not used because the frailty distribution is not identifiable from aggregate death counts, and it would quietly change the meaning of every age-specific rate.
- **Explicit competing-risks hazards with a copula.** The most complete treatment, allowing the two causes of death to be dependent; requires information on the degree of dependence that aggregate cross-sectional data cannot provide.

4 Cancer Onset: The Diagnosis Hazard

4.1 Functional form of the hazard

Mechanism. The annual probability that a cancer is clinically diagnosed at age a (given no earlier diagnosis) is a *piecewise-constant hazard on the log scale*:

$$\log h(a) = \beta_0 + \sum_{k=1}^K \beta_k \mathbf{1}\{a \geq t_k\},$$

with break-point ages t_k and coefficients β_k supplied by the configuration (the packaged default is $t = (-1, 0, 40, 50, 60, 70)$ with $\beta = (-11.9, 0.01, 0.28, -0.11, -0.1, -0.1)$). The value of $h(a)$ is capped at 1. The *diagnosis age* of an agent is then the first age a at which a Bernoulli trial with probability $h(a)$ succeeds, scanning ages from 0 upward; if no age succeeds, the agent is never diagnosed and receives a sentinel diagnosis age (110, the “unreal life length”) that means “no clinical cancer”. The scan is vectorised: for every agent, a column of uniform draws is compared with the hazard curve and the first success is taken.

Why this design. The functional form follows directly from what can be calibrated and what the model needs next.

(i) *The calibration target is an age-specific annual incidence rate*, and a piecewise-constant discrete hazard is exactly the quantity that corresponds to it: “the annual probability of a diagnosis given no diagnosis yet”. Choosing a hazard (rather than a distribution of diagnosis ages) means the fitting step is linear in the coefficients and can be solved in milliseconds, which is what makes an interactive web interface possible.

(ii) *The break points are placed where the observed curve changes slope.* Incidence rises steeply from mid-life, and the knots at 40, 50, 60 and 70 are the ages where the slope changes in the target data, so a handful of parameters captures the shape without imposing a functional family.

(iii) *The log scale does two jobs.* It keeps h strictly positive without constraints, and it lets the constant term express a baseline probability over several orders of magnitude — the fitted $\beta_0 \approx -11.9$ corresponds to a hazard of order 10^{-5} per year at young ages, which is the right order of magnitude for a rare cancer and cannot be represented comfortably on a linear scale.

(iv) *The cap at 1 and the first-success scan make the object a proper probability and a proper first-event.* Capping is required because the Bernoulli device needs a probability; taking the first success encodes the fact that “age at diagnosis” is a first-event time, so no agent can be diagnosed twice and no agent can accumulate two independent diagnoses.

Alternatives.

- **A smooth parametric hazard (Gompertz, Weibull, log-logistic, gamma).** One or two parameters, a smooth derivative, easy to communicate; the risk is systematic misfit precisely where it matters, at the turnover and in the late-life plateau of cancer incidence. It is the natural choice if the target is a single narrow age range rather than the whole lifespan.
- **Splines (natural cubic, B-spline, or a penalised P-spline).** More flexible than step functions and smoother than a piecewise-constant fit; costs extra parameters and a penalty weight to choose. This is the most natural refinement of the present design.
- **Using the observed age-specific rates directly as the hazard** (no fitting at all). Removes the identifiability question entirely; the cost is sampling noise, no smoothing, no ability to interpolate or extrapolate, and a subtle mismatch between an annual rate and an annual discrete hazard unless the person-years correction is applied.
- **A mechanistic carcinogenesis model** (two-stage clonal expansion / Moolgavkar–Venzon–Knudson, Armitage–Doll multi-stage, or multistage with genomic instability). Explains *why* incidence rises roughly as a power of age and links initiation and promotion to exposures, which would make risk-factor modelling possible. Not used because such models are not identifiable from age-specific incidence alone — many parameter combinations reproduce the same incidence curve — and because the extra structure does not improve the screening question being asked.
- **Age–period–cohort modelling.** The right tool for incidence *trends*; it needs several periods and cohorts, whereas this model deliberately simulates a single cross-section.
- **Separate hazards for progressive and non-progressive lesions** — the adenoma–carcinoma sequence in colorectal cancer, with a pre-clinical detectable phase of its own. This is the correct structural treatment of pre-cancer and is the single largest structural omission in the present model (Section 17); it requires lesion-level data.

4.2 What counts as “having cancer”, and the oldest agents

Mechanism. An agent is marked `has_cancer` if and only if the sampled diagnosis age is below the sentinel (110) *and* below the agent’s natural death age. A person who would die before their clinical diagnosis year is therefore never given a tumour history at all. In addition, agents who are already older than 98 at $t = 0$ are assigned the sentinel directly, so they can never develop cancer during the run.

Why this design. Both rules follow from the same decision: the model’s unit of cancer is *a cancer that would be clinically diagnosed in this person’s remaining lifetime*. That is exactly the event that produces the observed incidence and mortality, and exactly the event that screening could displace. An agent who dies before the diagnosis year contributes nothing to any reported quantity, so constructing a tumour history for them would add cost and a meaningless trajectory. The age guard has the same motive in a cruder form: the diagnosis-age scan is bounded by the maximum representable age, and back-calculating a preclinical onset age for an agent already past 98 would place the entire preclinical trajectory outside the simulated window, so the agents concerned are simply excluded.

Why this design. *What this costs, stated honestly.* The natural history is *diagnosis-driven*: everybody who has cancer in the model would also have been diagnosed clinically. The model therefore cannot represent a cancer that exists but never becomes clinically apparent — which is precisely the overdiagnosis mechanism. Without screening this is harmless, because such a cancer is invisible in every reported outcome anyway. Under a screening scenario it is not harmless: every screen-detected cancer is a “true” cancer that would have been diagnosed and could have killed, so in the presence of a real non-progressive fraction the model will overstate the benefit of screening. The age guard is also a blunt modelling shortcut with no epidemiological justification — it is a range guard dressed as a risk rule — and it should be revisited if the horizon, the initial age structure or the sentinel age changes.

Alternatives.

- **Onset-driven natural history.** Give every agent a preclinical onset (from an initiation hazard) and let progression, symptoms and screening determine whether and when it is ever diagnosed. Then a person can die with undiagnosed cancer, and both overdiagnosis and “screen-detected but would never have surfaced” become representable. This is the structurally correct design and the direction a serious upgrade should take; it requires a progression model and preclinical-detectability data that the packaged inputs do not contain.
- **A non-progressive (or slowly progressive) fraction.** Standard in screening models: a fraction `nc` of lesions never reach clinical significance. One extra parameter, and it is the accepted way to make overdiagnosis quantifiable — but it cannot be estimated from incidence and mortality data alone, so it would have to be imposed from the literature and made a target of sensitivity analysis.
- **No exclusion at all: build a tumour history for every agent regardless of its fate.** Conceptually uniform and it removes the special case, but it multiplies the

work for information that no output consumes, and it produces “cancer in people who are already dead”.

- **Removing the 98-age guard but widening the sentinel.** A pure implementation fix: the guard should either be dropped (letting the hazard decide) or replaced by an explicit, documented truncation age.

5 The Preclinical Phase and the Time-Scale Problem

5.1 Exponential lead time

Mechanism. For every agent with cancer the model draws a *lead time* (preclinical sojourn time, i.e. the time from the first malignant cell to clinical diagnosis) from an exponential distribution with mean `lead_time_mean` (3.5 years in the default configuration), so $t_{\text{ttd}} \sim \text{Exp}(\lambda)$, $\lambda = 1/\text{lead_time_mean}$. The onset age is then back-calculated from the *observed* diagnosis age: $\text{onset} = \max(\text{diag} - \lceil t_{\text{ttd}} \rceil, 0)$. The continuous value of t_{ttd} is used for the stage determination (Stage 6.3) and the integer value for the age arithmetic.

Why this design. The whole preclinical history is anchored to the diagnosis date rather than simulated forward from an onset date, and that choice is the reason the model can be calibrated from aggregate data at all.

(i) *The observable is the diagnosis age.* Anchoring the preclinical trajectory to it means the model reproduces the observed incidence exactly by construction and uses the lead time only to decide what the tumour looked like *before* a date that is already correct. Nothing has to be right about the absolute onset rate, so a whole class of misfit is removed.

(ii) *Exponential is the maximum-entropy choice with a given mean.* Assuming a constant progression rate — no memory of how long the tumour has already been present — is the least committal assumption available for a sojourn time, and it is exactly what the classical Markov screening literature assumes. That compatibility is not cosmetic: the relations between lead time, sojourn time and screening benefit derived in that literature remain usable to sanity-check the model’s behaviour.

(iii) *One parameter, because only one can be afforded.* The absolute scale of the preclinical phase is not identifiable (Section 5.2), so adding a second sojourn parameter would add a second unidentified quantity. The mean is the smallest sufficient summary: it is the scale of everything downstream (crossing times, incidence age, stage duration) and it is the single knob the sensitivity analysis turns.

(iv) *Memorylessness keeps the staging calibration closed-form.* Under an exponential lead time $P(\text{stage} \geq s) = \exp(-t_s/\mu)$, which is what makes it possible to *invert* the observed stage mix into crossing times with one logarithm per stage (Section 6.3) instead of solving a numerical calibration problem.

(v) *Continuous for the comparison, integer for the arithmetic.* Rounding the lead time before comparing it with the crossing times would bias the stage distribution (a rounding-down would systematically over-produce early stages), so the crossing comparison uses the continuous draw while the age arithmetic uses its ceiling.

Alternatives.

- **Gamma or Weibull sojourn times.** The standard generalisations, widely used in screen-detection models; exponential is their shape-1 special case. They buy a more realistic shape (few very long sojourns), but the shape parameter has to be assumed rather than estimated, so they trade one assumption for two.
- **A deterministic (fixed) lead time.** The simplest possible assumption, and the stage mix would then be a knife-edge function of a single threshold rather than a smooth probability; it also makes the preclinical duration unrealistically discrete.
- **Per-stage dwell times** (a sojourn time for each stage). Much closer to tumour biology and it makes stage duration directly interpretable; the available data cannot inform four separate dwell times, and it forces the crossing times onto an integer grid.
- **Size-driven detectability from the growth curve itself.** If the tumour is “detectable” once it crosses a size threshold, the sojourn time is whatever the Gompertz model says it is, which is beautifully self-consistent because the model already contains that curve. The missing inputs are a detectable-size threshold and a size-dependent test sensitivity, neither of which is in the packaged data.
- **Estimating the lead time from screening data** (screen-detected versus interval cancers, or a randomised screening trial). This is the only route to genuine identification, and it is the single highest-value data addition the model could receive.
- **Anchoring the mean to the literature** instead of leaving 3.5 as a free number, with the published range used as the sensitivity interval. Operationally identical to the present design, but it turns an assumption into a citable assumption — the cheapest available improvement.

5.2 Non-identifiability of the time scale, and what is fixed instead

Mechanism. The model does *not* estimate the preclinical time scale. It fixes the lead-time mean (`lead_time_mean`) and calibrates everything else and, separately, derives the stage-crossing times from the observed stage mix (Section 6.3). The fixed scale is then exposed as the single axis of the sensitivity analysis (Section 16).

Why this design. The reason is a genuine mathematical property of the system, not a shortage of effort. Let $c_s = C - \ln(K - \ln s)$ and write the crossing time of stage s as $t_s = c_s/B_{\text{ind}}$ (Section 6). Under the exponential lead time with mean μ ,

$$P(\text{diagnosed at stage} \geq s) = P(t_s \leq t_{\text{td}}) = \exp\left(-\frac{c_s}{B_{\text{ind}}\mu}\right),$$

so the observed stage mix constrains only the *product* $B_{\text{ind}}\mu$. Doubling the growth rate and halving the sojourn mean leaves every observable exactly as it was. Formally, the parameter vector (B_{ind}, μ) is unidentified along the hyperbola $B_{\text{ind}}\mu = \text{const}$ — a flat likelihood ridge, and a likelihood maximiser would happily return an arbitrary point on it.

There is a second, more basic reason. Onset — the appearance of the first malignant cell — is not an observable event. The lead time is therefore not a quantity that is measured badly; it is a quantity that is, in principle, unobservable in a clinical dataset, because diagnosis interrupts the very history one would want to observe. “Estimated lead time” in the literature always means “estimated under an explicit model”.

What the data *do* identify is: the diagnosis hazard (from age-specific incidence), the product $B_{\text{ind}}\mu$ (from the stage mix), and the terminal-phase hazard (from cancer mortality). The design choice is therefore to spend the one remaining degree of freedom as an *explicit assumption with an interpretable name* rather than as a fitted value that pretends to be an estimate. Fixing μ rather than B is the asymmetric but correct choice: μ has literature estimates for colorectal cancer and a plain clinical meaning (time from onset to diagnosis), whereas B has a scale that nothing outside the model can speak to.

Concretely, the consequence is a clean division of labour: μ is an assumption, the crossing times are *derived* from the data, and the growth-rate parameters are *fitted* to those derived times. So the model reproduces the observed stage mix by construction, and the single assumption is visible in the parameter file, in the document, and in the sensitivity plots.

Alternatives.

- **Report the lead-time range as the headline result, not as a sensitivity appendix.** Nothing about the maths changes; what changes is that the reader sees a family of curves as the answer rather than one curve plus a caveat. For a screening question this is arguably the more honest presentation, and it is a reporting change rather than a modelling change.
- **Bayesian treatment: a prior on μ , the rest fitted, and a full posterior.** The statistically cleanest way to say “ μ is a judgement” — the judgement enters as a prior, and its consequences arrive as credible intervals on every output rather than as a set of re-runs. Requires MCMC (or ABC) at a cost of hours instead of milliseconds, and it makes the headline numbers depend on a prior that is itself not derived from data.
- **Profile likelihood over μ .** Compute the fit as a function of μ and show the surface. Cheap (the model is fast), transparent, and it makes the ridge visible rather than merely described. This is the smallest possible step towards a proper uncertainty treatment.
- **Fix B instead of μ .** Symmetric and equally arbitrary, and strictly worse: the growth rate has no external anchor at all, so the choice would be uninterpretable and untestable.
- **Two-parameter sojourn time.** Worse, not better: it replaces one unidentified scale with two unidentified parameters, and no amount of prior opinion can identify them.
- **External identification from screening data.** A screening trial or a registry of screen-detected and interval cancers identifies the sojourn time directly; that is the only way to turn μ into an estimate. It requires individual-level screening data this model does not have.

6 Tumour Growth: The Reduced Gompertz Model

6.1 The growth curve

Mechanism. Tumour growth is described by the *reduced Gompertz* function

$$V(t) = \exp(K - \exp(C - Bt)),$$

where V is an abstract proxy for tumour mass (a “size”), t is years since the first malignant cell, K and C are shape parameters and B is the growth rate. The curve starts at $V(0) = \exp(K - \exp C)$ (a small tumour), saturates at $V(\infty) = \exp K$ (the carrying capacity), and has its inflection point at $t = C/B$, where the growth switches from accelerating to decelerating. Stages are the crossing times of the sizes 2, 3 and 4:

$$t_s = \frac{C - \ln(K - \ln s)}{B}, \quad s \in \{2, 3, 4\},$$

which is finite exactly when $\exp K > s$, i.e. when the carrying capacity is above the stage threshold. In the reference fit ($K = 1.451$, $C = 0.198$, $B = 0.550$, so $\exp K = 4.27$) the crossing times are 0.87, 2.26 and 5.34 years.

Why this design. Gompertz is chosen for four reasons, three substantive and one structural.

(i) *It is the classical tumour growth law and it matches the qualitative shape of the data.* Tumours grow roughly exponentially at first and then decelerate as they outstrip their blood supply and available space; the Gompertz curve has exactly that feature — exponential in the early phase, asymptotically flat late — and has been the default descriptive growth model in oncology for half a century.

(ii) *It is parsimonious without being rigid.* Two shape parameters (K, C) plus one rate (B) reproduce a wide range of growth curves, and each parameter has a plain meaning: $\exp K$ is a carrying capacity in size units, C fixes the initial size through $V(0) = \exp(K - \exp C)$, and B is the specific growth rate.

(iii) *Its crossing times have a closed form.* The inverse $t_s = (C - \ln(K - \ln s))/B$ is what makes the staging calibration a *linear-in-a-logarithm* problem and lets the model store each agent’s stage ages at generation time. A growth law whose stage times required numerical inversion would be both slower and much harder to reason about — and the closed form is also what makes the non-identifiability statement of Section 5.2 provable rather than empirical.

(iv) *Crossings of absolute sizes $\{2, 3, 4\}$ are a normalisation, not a restriction.* The model has three crossing times as calibration targets and three free parameters (K, C, B), so the fit is exactly identified: the threshold convention is absorbed by the fitted shapes. What the threshold spacing *does* affect is the individual-level dispersion of stages (through B_{std} , Section 6.2), because different tumours cross differently spaced thresholds at different times. That is a limitation to keep in view, not a hidden distortion of the population means.

Alternatives.

- **Exponential growth.** Simplest possible law; unbounded, so it has no carrying capacity, and it forces the crossing times to be equally spaced in log size — a strong restriction that the data would contradict.
- **Logistic growth.** Familiar and symmetric; the deceleration is symmetric about the inflection, which is known to be wrong for tumours (they grow for a long time before slowing), and its crossing times are symmetric for the same reason.
- **von Bertalanffy / Bertalanffy–Pütter.** Mechanistically motivated (growth as the balance of anabolic and catabolic terms) and in the same family as Gompertz; comparable behaviour, less standard in the screening literature.

- **Weibull or power-law growth.** Flexible; no carrying capacity, and no established clinical interpretation of the parameters.
- **A mechanistic, nutrient- or space-limited ODE model with explicit vasculature.** The most biologically grounded option; many parameters, and it needs volumetric growth data that aggregate cancer statistics do not contain.
- **Stochastic branching-process growth.** Models the clonal dynamics directly and — uniquely — can express *extinction* of early tumours, which is a genuine mechanism of non-progression. Computationally far heavier and not identifiable from the available data.
- **Piecewise-linear progression (no growth model).** A constant progression rate per tumour class, i.e. the classic multi-state Markov formulation. Simpler and directly fittable to stage data; it severs the link between growth and stage timing that makes the current model interpretable, and it leaves the notion of a “size” with nothing behind it.
- **Direct ordinal regression of stage on covariates** — the design this replaced. It reproduces the observed stage mix, but the stage becomes an arbitrary draw decoupled from any growth process, individual heterogeneity loses its physical meaning, and the failure is silent because the outputs still look plausible.

6.2 Individual heterogeneity in the growth rate

Mechanism. Each agent with cancer receives its own growth rate

$$B_{\text{ind}} = \exp(\ln B_{\text{pop}} + \varepsilon), \quad \varepsilon \sim \mathcal{N}(0, B_{\text{std}}^2),$$

so B_{ind} is log-normally distributed with median B_{pop} and multiplicative spread $\exp(B_{\text{std}})$ (a floor of 10^{-6} prevents a degenerate zero). The population-mean curve used for fitting uses B_{pop} without the random effect. B_{std} is *not* fitted: it is a hand-set value (0.1 by default, i.e. roughly $\pm 10\%$ multiplicative spread).

Why this design. Two arguments, one about realism and one about a specific screening phenomenon.

(i) *Growth rates really do vary, and the clinical fact is not a nuisance parameter.* Tumour doubling times for the same cancer differ between patients by factors, not percentages. Without heterogeneity, the stage is a deterministic function of the lead time: every patient with the same elapsed preclinical time would sit at the same stage, which imposes a rigid, artificial correlation between “how long the tumour has been growing” and “how advanced it is”. Heterogeneity decouples them, so two patients with identical lead times can legitimately be diagnosed at different stages.

(ii) *Length bias needs it.* Screening preferentially catches slowly growing tumours, because a tumour that spends longer in the detectable preclinical phase has more chances to be caught. This *length bias* — together with the associated lead-time bias — is one of the classical reasons screening trials overstate benefit, and it can only appear in a model in which growth rates differ between individuals. With $B_{\text{std}} = 0$ the phenomenon vanishes by construction and the model’s stage shift becomes too favourable and too clean.

Log-normal rather than normal is the natural parameterisation for a rate: positivity is automatic, the distribution is right-skewed (a few fast tumours, a floor of slow ones) as multiplicative quantities tend to be, and the two parameters have direct meaning —

B_{pop} is the median and B_{std} the log-scale spread. One subtlety worth remembering: the arithmetic mean is $B_{\text{pop}} \exp(B_{\text{std}}^2/2)$, slightly above the median, so a mean-based comparison against a single fitted rate carries a small, known offset.

Why this design. *An honest weakness.* B_{std} governs how much individual variation, stage dispersion and length bias the model produces — and it is set by hand, not estimated. Nothing in the current calibration constrains it, because the fitting step targets only the population-mean crossing times (K , C , B_{pop}). It is therefore a second unverified assumption alongside the lead-time mean, but unlike the lead time it is currently *not* varied in the sensitivity analysis. Turning it into a sensitivity axis is cheap and would materially strengthen the uncertainty statement.

Alternatives.

- **No heterogeneity** ($B_{\text{std}} = 0$). The simplest option and the one that makes the stage a deterministic function of the lead time. It removes length bias and stage dispersion entirely and makes the correlation between stage, lead time and survival artificially tight — so it should only be used as a deliberate sensitivity comparison, never as the baseline.
- **A Gamma-distributed growth rate.** Also positive, with independent control of shape and scale; less natural for a multiplicative quantity and no better suited to the data.
- **A discrete number of growth classes** (slow / medium / fast). Very easy to communicate and to vary in a scenario analysis; coarse, introduces artificial discreteness, and the class boundaries are arbitrary.
- **A normal (untruncated) growth rate.** Rejected: it admits negative rates, which are not merely unlikely but meaningless.
- **A full random-effects model of $\ln B$ with covariates** (age, sex, stage at onset). The right way to make growth heterogeneity explicit rather than incidental; needs covariate-specific growth information that the staging file does not contain.
- **Empirical volume-doubling times** from imaging series or from the literature. The only route to a defensible value of B_{std} ; it would convert a hand-set number into a cited one.

6.3 From growth to stage: staging as a derived quantity

Mechanism. For each agent with cancer the model computes its own crossing times t_2, t_3, t_4 (from B_{ind} , Section 6) and compares them with the continuous lead time: the stage at clinical diagnosis is 4 if $t_4 \leq t_{\text{td}}$, else 3 if $t_3 \leq t_{\text{td}}$, else 2 if $t_2 \leq t_{\text{td}}$, else 1. In other words the stage is *the number of size thresholds the tumour has crossed by the time it is clinically diagnosed*. The full stage-age array stored per agent is $[\text{onset}, \text{onset} + t_2, \text{onset} + t_3, \text{onset} + t_4, \text{onset} + t_4 + \delta]$, where δ is an exponential draw for the terminal phase after the last crossing.

The crossing times used as calibration *targets* are obtained by inverting the observed stage mix. Because the lead time is exponential, $P(\text{stage} \geq s) = \exp(-t_s/\mu)$, so

$$t_2 = -\mu \ln(1 - p_1), \quad t_3 = -\mu \ln(p_3 + p_4), \quad t_4 = -\mu \ln p_4,$$

where p_s is the observed proportion at stage s and μ is the lead-time mean. For the USA colorectal staging file the observed mix is 0.220/0.256/0.308/0.218, which yields the targets 0.87/2.26/5.34 years; the resulting fit reproduces the mix with 0.225/0.250/0.284/0.241 ($N = 20\,000$, seed 2022).

Why this design. Staging is derived rather than drawn, and the difference matters more than it looks.

(i) *It makes the stage mix a consequence instead of an input.* When the stage is drawn from a distribution, reproducing the observed mix is a tautology — the model was told the answer. Here the mix is produced by the interaction of two physical quantities (how long the tumour has been present, how fast it grows), so agreement with the observed mix is a genuine, falsifiable calibration check. The residual gap (0.225/0.250/0.284/0.241 versus 0.220/0.256/0.308/0.218) is small and attributable to sampling noise, to B_{std} , and to the interaction between the stage and the age-dependent diagnosis hazard — not to a systematic misfit.

(ii) *It is what makes the per-agent growth rate meaningful.* If the stage came from a regression, B_{ind} would be a number with no downstream consequence; because the crossing times are computed from it, the growth rate controls the stage, the stage ages, and therefore the timing of the terminal phase. Growth heterogeneity has observable consequences, which is exactly what a structural parameter should have.

(iii) *The inversion is closed-form, so the calibration target is data-derived and auditable.* One logarithm per stage turns the staging file into three numbers that can be printed, checked by hand, and traced back to the source table. There is no inner optimisation loop to hide a mistake in.

(iv) *It respects the non-identifiability accounting.* The inversion consumes μ (the assumed scale) and returns the crossing times; the Gompertz fit then consumes the crossing times and returns K , C , B_{pop} . Three targets, three parameters — an exactly identified chain in which the single assumption is visible at the front.

Alternatives.

- **Draw the stage directly from the observed distribution.** Reproduces the mix exactly, at the cost of making the stage exogenous: growth heterogeneity, lead time and screening can no longer shift the stage except by an explicit, arbitrary fudge, so the tumour growth model becomes decorative.
- **Multinomial or ordinal regression of stage on age, sex and other covariates.** The design this replaced. It fits the mix well and is easy to implement, but it decouples the stage from the growth process, leaves the individual growth rate meaningless, and its coefficient interpretation (“the effect of age on the log-odds of being at a later stage”) has no mechanism behind it.
- **A latent stage-progression process with transition rates** (the standard multi-state formulation). Clean, and it makes dwell times explicit; it needs observed stage-transition data (e.g. from serial screening rounds), which the packaged inputs lack.
- **Clinical TNM staging** (tumour size T , nodal involvement N , distant metastasis M). The clinically correct and far more informative description, and it would allow a genuine screening stage shift to be measured against real thresholds; it requires

the size and nodal information that a stage-only staging file does not contain.

- **Discrete growth classes with fixed dwell times** (slow/intermediate/fast tumours, each with a fixed dwell time per stage). Simple and transparent, and it is how several published screening models are parameterised; it makes the dispersion of stages a discrete choice rather than a continuous one.

7 Aggressiveness: A Cure-Modifying Tumour Label

7.1 What the label is and what it does

Mechanism. Every cancer agent is labelled *aggressive* or not by a Bernoulli draw whose probability is taken from the staging data: for a tumour diagnosed at stage s the probability is the observed proportion of aggressive tumours in that stage, computed from the file’s **Aggressiveness** column (USA colorectal data: 0.096/0.198/0.437/0.639 for stages I–IV, i.e. aggressiveness and stage advance together). The label then affects *exactly one thing* — the probability of cure. The baseline cure probability p is converted to odds, multiplied by a per-stage odds ratio (**aggressiveness_cure_odds_ratio**, default 0.667 for every stage), and converted back:

$$\text{odds}' = \frac{p}{1-p} \times \text{OR}_s, \quad p' = \frac{\text{odds}'}{1 + \text{odds}'}.$$

A value of $\text{OR}_s = 1$ is a strict no-op (the transform is skipped), so the assumption can be switched off without touching any other behaviour.

Why this design. Four reasons, and they interlock.

(i) *It is the only phenotype dimension for which the model has data.* The staging file carries an aggressiveness flag, so the per-stage prevalence is *measured*, not assumed. Ignoring it would throw away real information; inventing a different tumour-phenotype axis would have no data behind it at all.

(ii) *It is deliberately not a growth modifier, to avoid double counting.* Aggressive tumours are, in clinical intuition, fast-growing ones — so the tempting design is to let the label change B . That would confound it with B_{std} , which already carries growth heterogeneity, and the calibrated stage mix would then have to absorb two overlapping sources of variation. Assigning the label to the cure probability puts it where it is observable: in survival, not in staging.

(iii) *The odds-ratio form is the right algebra for a cure probability.* Cure is a binary outcome, so an odds ratio is the standard effect measure; it composes multiplicatively with the baseline odds, cannot leave $[0, 1]$, is symmetric in framing (OR and $1/\text{OR}$ express “harder” and “easier” with the same parameter), and reduces to the identity at $\text{OR} = 1$.

(iv) *The direction is a clinical prior, and the form leaves room for stage-specific evidence.* That an aggressive tumour is harder to cure is a reasonable prior, and $\text{OR} < 1$ encodes it. The parameter is a vector over stages even though the default sets all four equal, so a user with stage-specific treatment-response evidence can enter it without changing the design.

Why this design. *Two limitations to state plainly.* First, the odds ratio itself is *not estimated from data* — it is set by hand. The staging file contains no treatment-outcome column, so there is nothing in the packaged inputs that could identify it: what is measurable (the prevalence by stage) is measured, and what is not (the effect of the label on cure) is an explicit assumption with a clinical prior. Second, because aggressiveness is strongly associated with stage (0.096 versus 0.639), a pooled comparison of aggressive against non-aggressive survival confounds the odds ratio with the stage mix — the aggregate numbers will show a large survival gap that is mostly a stage effect. This is a Simpson’s-paradox trap, and the model avoids the trap by reporting survival curves and cure fractions *stratified by stage and aggressiveness* rather than only pooled (Section 11.2).

Alternatives.

- **Aggressiveness affecting growth (B).** Biologically natural, and it would make the label do something visible in the staging data. Rejected because it double-counts against B_{std} and destroys the exact stage-mix calibration.
- **A continuous latent aggressiveness score.** More realistic — aggressiveness is a spectrum, not a flag — and it would let the cure odds vary smoothly; it cannot be derived from a binary column, and it would add a non-identifiable latent dimension.
- **A mixture cure model with the label as a covariate.** Statistically the standard formulation, and the present design is effectively its logistic special case; a full mixture model would estimate the uncured fraction directly, at the cost of a likelihood that needs treatment follow-up data.
- **Subtype-specific (grade, MSI/MMR status, molecular subtype) survival from the literature.** Substantially more informative and clinically meaningful than a binary flag, and it would make the cure effect evidence-based. Requires subtype-level data the aggregate files do not contain.
- **Making aggressiveness modify the cancer-death hazard instead of the cure probability.** Equivalent in its effect on mortality, different in mechanism (it changes survival given non-cure rather than the chance of cure). Arguably more direct; harder to calibrate and less interpretable as “treatment success”.
- **Dropping the label.** Simplest, and it removes a hand-set parameter — but it discards the only tumour-phenotype information the inputs provide.

8 Treatment and Cure

8.1 The cure probability

Mechanism. At clinical diagnosis the model computes a cure probability

$$p_{\text{cure}} = \min\left(1, \alpha(\text{diag}) \times \theta_s\right),$$

where θ_s is the stage-specific treatment efficiency and $\alpha(\cdot)$ is an age-dependent efficiency factor. $\theta = [0.7, 0.3, 0.2, 0.2]$ for stages I–IV. α is a step function of the age at diagnosis with break points 40/50/60/70 and values $[1.0, 0.9, 0.5, 0.3, 0.1]$. If the tumour is aggressive, the cure *odds* are multiplied by the per-stage odds ratio of Section 7 before converting back. Cure is then a Bernoulli draw with the resulting probability. In the reference configuration p_{cure} spans 0.70 (stage I, diagnosed young) down to 0.02 (stage IV, diagnosed after 70).

Why this design. The cure probability is factored into stage and age because those are the two axes on which curability is actually observed and reported.

(i) *Stage dominates and the data measure it.* Stage at diagnosis is the strongest single predictor of cancer survival in every registry, and it is the dimension along which the staging file is organised. Making it the primary factor aligns the model with the observable that constrains it.

(ii) *Age enters as a step function deliberately.* The age gradient in cure is real — older patients tolerate treatment worse, are treated less aggressively and carry more competing mortality — but a smooth age effect would need shape parameters that nothing in the inputs can identify. A four-step function is the granularity at which cancer statistics are published and at which the outputs are reported, it needs no shape assumption, and it is directly legible to a clinician. The step values read as “relative cure efficiency in this age band”.

(iii) *Multiplication is the right composition.* It keeps the result inside $[0, 1]$ without clamping except at the top, it makes each factor interpretable as a relative multiplier (a patient with the best possible age efficiency receives exactly the stage-specific efficiency), and it makes switching an effect off a one-line change (all age constants set to 1).

(iv) *The cap at 1 is a correctness requirement* — the Bernoulli device needs a probability — and it is harmless because the reference values never approach it.

(v) *The odds-ratio transform for the aggressive label behaves sensibly.* It is monotone and bounded, so the label’s effect is largest at intermediate cure probabilities and shrinks towards zero at both extremes, where a cure probability is already nearly certain or nearly hopeless.

8.2 What “cured” means for death

Mechanism. A cured patient receives a cured death age, but the decision of *when* the person dies is made by the competing-risk rule of Section 10: the agent leaves the population at its natural death age, and the cause is attributed in the statistics layer. Each year the model therefore compares two stored ages and counts a cancer death when the agent has cancer, is not cured (neither clinically nor by screening), the untreated cancer-death age equals the current age, and that age does not exceed the natural death age. The uncured patient’s cancer-death age is the last stage crossing plus an exponential draw for the terminal phase, $\delta \sim \text{Exp}(\lambda_{\text{terminal}})$, where $\lambda_{\text{terminal}}$ is fitted to the observed cancer mortality (0.35 per year in the reference fit, i.e. a mean terminal survival of about 2.9 years).

Why this design. Defining cure as “the cancer no longer determines either the date or the cause of death” is what keeps the screening arithmetic exact and honest.

(i) *The paired counterfactual stays within one agent.* Because the untreated cancer-death age is stored for every patient, a cure is simply a change of which cause the statistics count. The lives-saved and years-saved outputs are then genuinely within-individual differences, not differences between two populations with different sampling noise — which is what makes them meaningful at the sample sizes used.

(ii) *No post-cure survival model is needed.* A cured patient’s remaining lifetime is the population’s remaining lifetime, which is precisely the competing-risk assumption already

made for everyone else. Introducing anything else would require a cure-specific mortality (treatment toxicity, late relapse) and therefore data.

(iii) *The terminal phase carries the timing of cancer death, and it is the one treatment-adjacent quantity that is actually fitted.* Cancer mortality in the data is dominated by how long uncured patients survive after diagnosis, so a single fitted hazard for the terminal phase is the most economical way to match that observable.

Why this design. *The confound, restated because it matters.* Because the cure probabilities are hand-set and only the terminal hazard is fitted, the observed cancer mortality constrains the *product* of the two rather than each separately. A scenario with more cures and faster cancer death can reproduce the same mortality curve as one with fewer cures and slower death. Consequences: cure fractions reported by the model should be read as scenario assumptions, not as estimates; the fitted terminal hazard absorbs whatever the assumed cure probabilities get wrong; and only stage-specific relative survival with follow-up could separate them. This is the second flat ridge in the model, and it deserves the same prominence as the lead-time ridge of Section 5.2.

Alternatives.

- **Stage-specific relative-survival tables used directly** (the classical approach in cancer cost-effectiveness models). The observable becomes the input, so no cure mechanism is needed and published survival is reproduced exactly; the mechanism is lost, “why” a patient survives longer cannot be asked, and treatment advances can only be represented by swapping the table.
- **A mixture cure model with the cure fraction estimated from follow-up data.** Statistically the standard formulation, and the current design is essentially its logistic core; it requires individual follow-up and would identify the cure fractions that are currently assumed.
- **Excess-mortality (relative survival) hazard modelling.** Model the cancer-attributable excess hazard as a function of stage, age and time since diagnosis instead of a probability of cure. More faithful to how registry data behave, and it avoids the binary “cured / not cured” abstraction; it needs population life tables by age, sex and period, and it makes the cured flag — and hence the lives-saved accounting — only approximate.
- **Treatment-pathway modelling** (surgery, chemotherapy, radiotherapy, each with its own outcome distribution). Clinically the most informative, and the natural home of several currently inactive parameters (complications, treatment mortality); it needs treatment data that the packaged inputs lack.
- **Post-cure mortality adjustment** (a multiplier on the natural-death hazard for cured patients, representing treatment toxicity or late relapse). A small, honest extension; it needs data to set the multiplier and would break the “cured patients are ordinary people” simplicity.
- **A strict cure that deletes the cancer death hazard entirely** and re-draws other-cause mortality from a cause-deleted life table. Internally the purest formulation; it requires the cause-deleted life table discussed in Section 3.

9 Screening: Schedule, Participation, Test Performance

9.1 Invitation schedule and participation

Mechanism. A screening round takes place in simulated year y if $y \geq \text{screening_date}$ and $(y - \text{screening_date})$ is a multiple of `frequency` (defaults: the programme starts in simulated year 10 and runs every 2 years). In a round, an agent is *eligible* if it is alive, its age lies in `[start_age, finish_age]` (50–60 by default), it has not already been screen-detected, and it has not yet been clinically diagnosed (current age below its diagnosis age). Every eligible agent takes the test with probability `participation_rate` (0.5), drawn independently in each round.

Why this design. The schedule is expressed as an age window plus a period because that is how screening programmes are actually specified and evaluated, and it is representable exactly on the annual time grid (Section 2.2). Two eligibility rules carry real modelling content. Excluding already-detected agents keeps the history of an agent’s screening path simple and reflects the clinical reality that a person found to have cancer leaves the screening pathway. Excluding agents already past their clinical diagnosis age guarantees that a screen-detected cancer really is a cancer caught *earlier* — which is the only mechanism by which screening can help in this model — and it prevents a person from being counted both as screen-detected and as clinically diagnosed.

Why this design. Independent per-round participation is the maximum-entropy assumption given a single aggregate attendance rate, and it is the only one that an aggregate rate can support. It also reproduces the realistic phenomenon of intermittent attendance: a person may attend at 52 and skip 54. What it *cannot* represent is the well-documented subgroup of never-attenders (and the correlated over-attenders). That is a genuine simplification: real programmes’ benefit depends on reaching the under-screened, and a model with independent attendance will overstate the population-level effect of a given average participation rate.

9.2 Test performance

Mechanism. The test is described by a true-positive rate and a false-positive rate, chosen as a pair from a menu indexed by `selected_test`: $\{50, 60, 70, 80\}$ map to $TP = \{0.6, 0.7, 0.9, 0.95\}$ and $FP = \{0.05, 0.07, 0.1, 0.15\}$. Only the true-positive rate is actually applied: an agent with preclinical cancer tests positive with probability TP , and an agent without preclinical cancer tests positive with probability $0.1 \times TP$.

Why this design. A menu of (TP, FP) operating points makes the accuracy trade-off explicit and prevents the physically impossible combination of a perfect test: the user can only pick from the efficient frontier that the menu offers. Expressing the test as TP/FP applied to the two subgroups (with versus without preclinical cancer) is the classical 2×2 formulation and needs no distributional assumption about test values — which matters because no test-value data are shipped.

Why this design. *Two documented inconsistencies in this area, both of which should be fixed rather than explained.* (i) The configured false-positive rate is inert: the false-positive probability used in the simulation is $0.1 \times \text{TP}$, while `test_fp` is read, stored, exposed in the interface and never applied. The result is that changing the test in the menu changes the false-positive rate only through its correlation with TP, and always at the fixed 10% ratio — so the menu’s FP column is decorative. (ii) The aggressiveness cure odds ratio (Section 7) is applied on the clinical cure path but *not* on the screening cure path, so an aggressive tumour detected by screening is, in that one respect, treated as if it were not aggressive. Both are small in magnitude and neither breaks the model, but they mean the model’s screening module does not fully honour its own parameter set.

9.3 What screen detection does

Mechanism. If an eligible, participating agent has preclinical cancer (it has cancer and its onset age has passed), a positive test marks it screen-detected, records the screening age, and determines the stage at detection by counting how many crossing ages have already passed at that moment (the same derived-stage device as at clinical diagnosis, Section 6.3). A cure is then drawn with the *same* cure model as clinical diagnosis — the age-efficiency factor times the stage-specific treatment efficiency — but with an independent random draw. If the draw cures, the patient is flagged screening-cured and receives a screening death age no earlier than five to ten years after detection; if it does not cure, the clinical-cure flag is *cleared* and the untreated terminal age stands.

Why this design. The single most important design decision here is that screen detection re-uses the clinical cure model with a fresh draw, and that the cure probability depends on the stage *at which the cancer was found*. That is precisely where screening benefit comes from in this model: earlier detection means an earlier stage, and an earlier stage means a higher cure probability (0.7 at stage I versus 0.2 at stage IV). No separate “screening is better” effect is assumed — the benefit is entirely a stage-shift consequence, which is the honest way to model it and the way that is falsifiable against trial results.

Why this design. *The override, and why it must be stated.* Screen detection *replaces* the clinical trajectory: a failed screening cure clears the clinical-cure flag, so a patient who would have been cured clinically can end up dying of cancer once detected by screening. The paired lives-saved accounting is protected against double counting — an agent is counted as saved only if it is screening-cured *and* was not clinically cured — but the override means the model does not treat “the screen-detected patient who would have done fine anyway” as a neutral case. The reported screening effect is therefore the net of a benefit channel (earlier detection, lower stage, more cures) and a harm channel (overriding a favourable clinical outcome). In the reference run only 5 patients are screening-cured against 22 screen-detected, which is consistent with the harm channel being real rather than a corner case.

9.4 False positives

Mechanism. A participating agent without preclinical cancer tests positive with probability $0.1 \times \text{TP}$; the event is counted per year and contributes to the reported false-positive

curve. It has no other consequence: no follow-up procedure, no work-up cost, no anxiety, no effect on future participation.

Why this design. The model counts what it can count. False positives matter to a screening programme mainly through the burden of unnecessary follow-up, its costs, and the possibility of behavioural change — and the packaged inputs contain information about none of these. Counting them keeps the phenomenon visible in the results, and makes the rate’s inertness (previous subsection) obvious to anyone who varies the test menu, without inventing consequences. Deliberately not modelling a known effect is defensible; silently modelling it with invented numbers is not.

Alternatives.

- **A continuous-time screening process with exact instants.** Removes the discretisation of screening ages and lets detection happen the moment a tumour becomes detectable; requires an explicit detectability threshold, which the data do not provide, and it gives up the clean annual grid.
- **Correlated (“sticky”) participation, never-attenders, invitation-response heterogeneity.** A well-supported and important refinement: screening benefit is concentrated in attenders, and a programme with 50% attendance by a random half behaves differently from one with 100% attendance by half the population. Needs individual attendance data; the current design cannot express it.
- **Sensitivity and specificity instead of TP/FP.** Mathematically equivalent to what is applied ($\text{sensitivity} = \text{TP}$, $\text{specificity} = 1 - \text{FP}$); the present framing is chosen because the model applies the test to an agent known to have preclinical cancer, which is the definition of sensitivity.
- **A size-dependent test sensitivity** (the test improves as the tumour grows). What real tests do, and it strengthens the length bias; it requires a link between the Gompertz “size” and detectability, plus detection-probability-by-size data — a natural companion to the size-threshold alternative of Section 5.2.
- **Modelling the downstream consequences of a false positive** (follow-up procedures, their costs and complications, anxiety, reduced future attendance). The right long-run direction; it needs health-economic and behavioural data that are absent.
- **Route-specific treatment realism.** The present assumption is that a screen-detected cancer and a clinically detected cancer of the same stage are treated identically. Real screen-detected tumours often have more favourable biology, so this may understate the benefit; the alternatives are a route-specific cure parameter or route-specific survival from a trial.
- **An explicit interval-cancer channel** (cancers surfacing between rounds). It arises here endogenously, because the diagnosis hazard keeps operating between rounds; making it explicit means measuring it, which trial data would allow.
- **Programme scale-up over time** rather than switching on in year 10. A scenario and reporting question rather than a structural one.

10 Death Attribution and the Paired Comparison

Mechanism. The model never *draws* a cause of death: it *derives* one, each year, from the ages and flags it already stores. The categories are: 0 = no death recorded (alive), 1 = non-cancer (natural) death, including a cancer patient whose natural death age arrives first, 2 = cancer death on the untreated natural-history path, 3 = death averted by screening (the patient would have died of cancer but was cured by the screen), 4 = death from other causes after a clinical cure.

Two mortality series are produced by the *same* agents in a *single* run: the untreated series, using each patient’s natural-history cancer-death age, and the screening-adjusted series, using the effective death age (the screening death age for screen-detected, screen-cured patients, and the untreated age otherwise). Screening benefit is reported as *lives saved* — patients who are screening-cured, were not clinically cured, and whose untreated cancer death would have fallen inside the simulated window (3 people in the reference run) — and *years of life saved*, the accumulated difference between the two death ages (14.4 years in the reference run, about 4.8 years per person saved).

Why this design. (i) *Deriving the cause makes the categories mutually exclusive and exhaustive by construction.* Every agent falls into exactly one category in every year, so the counts add up and no separate bookkeeping can drift out of step.

(ii) *Both arms come from the same agents, so the comparison is paired.* This is the single largest statistical-efficiency decision in the model. Because the untreated trajectory is already stored for every patient, the screening effect is a within-individual difference, and the variance of that difference is orders of magnitude smaller than the variance of a difference between two independently simulated populations. A screening benefit of a few lives per 20 000 agents is measurable at all precisely because of this design; with two independent arms it would be buried in sampling noise.

(iii) *No re-simulation means no seed artefacts in the contrast.* The counterfactual is read from a stored number rather than obtained by running a control scenario, so the contrast does not inherit the control run’s randomness and cannot be tuned by choosing a seed.

(iv) *The “not clinically cured” clause is the correct counterfactual.* An agent counts as saved only if screening cured it *and* it was not going to be cured anyway. Without that clause, every early detection would count as a success, including the patients who would have been cured regardless — which is exactly the error that makes naïve screening evaluations overstate benefit.

Why this design. *Three honest qualifications.* (a) Agents leave the population only at their natural death age, so a recorded cancer death does not remove the agent (§3.2): the age structure follows the all-cause life table and cause-specific rates use the general-population denominator. This is the standard cause-specific device under an independence assumption, and it is a natural target for validation. (b) One branch of the saved accounting tests whether the natural death precedes the untreated cancer death, i.e. whether the person would *not* have died of cancer. It is empty in the reference run, and it is structurally unable to contribute years (because the screening death age is never earlier than the untreated one), so it can only inflate the people counter. It is a latent inconsistency rather than a present error, and it should be tightened. (c) Restricting lives saved to deaths that would have occurred inside the simulated window makes the reported figure

a *within-horizon* number, not a lifetime number: a person saved whose untreated death would have fallen in year 45 of a 30-year run is not counted. That understates lifetime benefit and makes the headline figure depend on the horizon — worth stating explicitly wherever it is quoted.

Alternatives.

- **Two independent populations (a screened and an unscreened cohort).** Simpler to implement and easier to explain; the contrast then carries the full Monte Carlo variance, so the same precision needs far more agents, and the answer becomes sensitive to the pair of seeds chosen. It also cannot express “this same person’s two possible lives”.
- **A same-seed twin run with screening disabled.** Recovers a paired contrast without storing the untreated ages, at the cost of doubling the runtime and of relying on the seeds aligning exactly. The current design obtains the same information for free.
- **Excess-mortality / relative-survival attribution** (cancer deaths as the excess over a background life table, with attributable fractions). Standard in registry analysis and it does not need a modelled cure mechanism; it abandons the exact paired accounting, needs cause-deleted background mortality, and makes “saved” a probability rather than a count.
- **Probability-of-causation attribution** (a death is partly attributable to each cause). More faithful to how deaths are actually certified; it destroys the clean integer counts that the cost and benefit outputs are built on.
- **Lifetime accounting with survival extrapolation** beyond the horizon. Gives the genuinely comparable “lives saved” figure used in trial reporting; needs extrapolation beyond the data and a discount rate for the years.
- **QALY- or DALY-weighted outcomes** instead of raw lives and life-years. The right currency for policy comparison; needs utility weights, a discount rate and a perspective (health system versus societal) — none of which the packaged inputs contain.

11 Reported Outcomes

11.1 Counting, rates and their denominators

Mechanism. All outputs are accumulated as counts indexed by *year* and *age*: the age distribution, the number at risk, incident cases (agents reaching their onset age), diagnoses (agents reaching their diagnosis age), and cancer deaths in both the untreated and the screening-adjusted series. A rate at age a is then the total number of events at that age divided by the total person-years at risk at that age, i.e. an average annual rate over the horizon. Two deliberately different incidence-type curves are produced: *incidence* counts the (unobservable) preclinical onset, and *diagnosis* counts the (observable) clinical detection.

Why this design. Counting on the same grid as the data means the model’s output can be compared with the published tables used to calibrate it without any re-binning,

which is what makes calibration and validation audits possible at all. Splitting the two incidence-type curves is a small design decision with a large diagnostic payoff: because the onset curve is a modelled quantity and the diagnosis curve is a calibrated one, their divergence is exactly the assumed lead time made visible. Anyone can therefore see how much of the model’s output depends on the unobservable quantity — which is the whole point of the lead-time sensitivity analysis.

Why this design. *The denominator is the general population, not the cancer-free population.* Because agents are removed only by natural death (§3.2), the at-risk denominator for cause-specific rates includes people with a recorded cancer death. This is the standard cause-specific construction under the independence assumption; it is also why the reported rates should not be interpreted as “probability of cancer death among cancer patients”. Crude rates by age are reported as well, so a user comparing two populations must age-standardise before drawing conclusions.

Alternatives.

- **Age-standardised rates** (direct standardisation to a reference population) instead of crude rates. Necessary whenever two datasets with different age structures are compared — which is the normal case here, since the packaged datasets differ by country. Computing the standardisation in the model rather than leaving it to the reader is a small, high-value improvement.
- **Cumulative risk / lifetime risk** (“the probability of a diagnosis before age 75”). The headline metric in most cancer statistics, and more interpretable to a non-specialist than an annual rate; it is a re-expression of what the model already computes.
- **Incidence-based versus mortality-based outcome framing.** The epidemiological convention is to frame results either in terms of cases avoided or deaths averted; the model reports both, which is the superset, but the choice matters when a downstream decision model expects one specific currency.
- **Interval-based rather than single-year counts** (five-year age bands, calendar periods). Aligns with how statistics are published and reduces the sparsity of small counts; loses resolution exactly where the model is most informative (the screening window).
- **Rate smoothing** (as is applied to the sensitivity curves). Makes charts readable but changes the numbers; it should stay a presentation-layer choice, as it is at present.

11.2 Survival, stratified by stage and aggressiveness

Mechanism. Two overall cause-specific survival curves are produced — one on the untreated series and one on the screening-adjusted series — both measured from diagnosis. In addition, and more recently, the model reports cause-specific survival from diagnosis *stratified by stage and by aggressiveness* (each of stages I–IV plus the pooled curve, in both the aggressive and the non-aggressive subgroup), together with the *cure fraction* of every one of those strata. The event is defined precisely: a patient dies of cancer if it is not cured (clinically or by screening) and its untreated cancer-death age is reached before its natural death age. Cured patients, and uncured patients who die of something else

first, are censored. The curve is accumulated as a discrete hazard, $H_t = H_{t-1} + d_t/n_t$ for event counts d_t and numbers at risk n_t , with $S_t = \exp(-H_t)$ over a 30-year horizon. In the reference run the pooled cure fraction is 13.1% for aggressive and 27.2% for non-aggressive patients.

Why this design. (i) *Cause-specific rather than all-cause* because the quantity of interest is the cancer’s effect, and because all-cause survival mixes in the larger non-cancer mortality — which differs between the packaged datasets, so an all-cause curve would make cross-dataset comparison meaningless.

(ii) *Measured from diagnosis* because that is the clinically meaningful time origin and the one used in registry survival reporting, so the output is directly comparable with published figures.

(iii) *The “cured $\Rightarrow$ censored” convention is the correct cause-specific device.* Once a patient is cured the cancer can no longer kill, so for a cancer-specific curve that patient is no longer at risk of the event. Counting a cured patient as an event would be plainly wrong; treating it as censored is the standard construction, and it is the same rule that the death-attribution layer uses (§10), so the two cannot disagree.

(iv) *Stratification is the Simpson’s-paradox guard.* Because aggressiveness rises steeply with stage (0.096 at stage I to 0.639 at stage IV, §7), the pooled aggressive-versus-non-aggressive curves are mostly measuring stage composition. The per-stage comparison is the only one that isolates the odds-ratio assumption, so reporting pooled curves alone would invite exactly the wrong conclusion.

(v) *Reporting cure fractions alongside the curves makes the central mechanism auditable.* The model’s benefit channel is “earlier stage $\rightarrow$ higher cure probability”, so the reader should be able to see how many patients the cure abstraction actually rescues, by stage and by aggressiveness — and to check that the reported fractions move in the direction the parameter implies.

Why this design. *The caveat that follows from §8.* These curves are conditional on the assumed cure probabilities, which are hand-set and confounded with the fitted terminal hazard. A survival curve can therefore be “right” for partly compensating reasons. The honest reading is: the curves are a diagnostic of the model’s internal mechanism and a tool for scenario comparison, not an independent estimate of cancer survival. Comparing them against externally published stage-specific survival is the test that would settle it.

Alternatives.

- **All-cause survival.** Unambiguous — it needs no judgement about the cause of death — but confounded by background mortality, so it cannot be compared across populations with different life tables.
- **Relative / net survival** (observed survival divided by expected survival from a background life table). The registry standard and the only construction that makes the model’s survival comparable with published figures; it needs background mortality by age, sex and period.
- **Excess-hazard modelling of survival** (the cancer-attributable hazard as a function of stage, age and time since diagnosis). Closer to how the data behave and it removes the need for a binary cure; it requires individual follow-up to estimate.

- **A fitted parametric survival model** (Weibull, Gompertz, log-normal) instead of a non-parametric curve. Compact summary statistics, extrapolation, and easy comparison of models; it adds a shape assumption that the data cannot always support.
- **Conditional survival** (survival given that five years have already been survived). Clinically the most interpretable framing of the “cure plateau”, and it makes the cure abstraction visible without adding a parameter.
- **Hazard ratios between the arms** instead of full curves. A compact effect measure that is easy to tabulate; it discards the time pattern, which is the aspect screening most affects.
- **Stratification by additional axes** (age band, route of detection, dataset). Cheap to compute, and it would expose interactions the pooled reporting hides; the risk is small strata and noisy curves, which is why the horizontal resolution should stay at stage and aggressiveness.

11.3 Stage shift: why the two stage distributions are not comparable

Mechanism. The model reports the stage distribution of clinically diagnosed cancers and, separately, of screen-detected cancers. In the reference run the clinical mix is 0.225/0.250/0.284/0.241 (mean stage 2.55), while the screen-detected mix ($n = 22$) is 0.000/0.318/0.500/0.182 (mean stage 2.86). For those same 22 agents, however, the screening stage *minus* the clinical stage averages -0.68 : 13 of 22 were detected at an earlier stage (three by two stages, ten by one), eight at the same stage and one later, with a mean of 4.8 years between detection and the clinical diagnosis that would otherwise have occurred.

Why this design. These numbers must be read together, because the two stage distributions are *not like-for-like* and reporting only the marginals would invite the wrong conclusion in either direction. Screening can only catch a person who happens to be in the preclinical phase at a round, and a tumour with a long sojourn has more opportunities to be caught. That is length bias: the screen-detected group is enriched with slowly growing tumours, which have had *more* time to progress. It is therefore entirely possible — and it happens in the reference run — for the screen-detected mix to look worse than the clinical mix in cross-section, while every within-person comparison shows that the screened tumour was found earlier than it otherwise would have been.

The model’s benefit mechanism is precisely the within-person shift: a lower stage at detection implies a higher cure probability (Section 7, stage I 0.7 versus stage IV 0.2). Measuring the mechanism therefore requires comparing each detected agent with its own counterfactual, which the model can do because it stores the clinical stage for every agent. The mean detection-to-clinical-diagnosis interval — about 4.8 years here — is the quantity that caps the achievable benefit: it is how much earlier the screening route acts.

Why this design. *Sample-size warning.* Twenty-two screen-detected cancers in 20 000 agents is a small count, so the screen-detected stage mix carries wide uncertainty and should not be quoted as a point estimate without an interval or an ensemble over seeds. The within-person mean shift is more stable because it is a paired quantity, which is

the same reason the paired design was chosen for the benefit accounting (§10). This is also a practical argument for the standardised, automated sensitivity sweep described in Section 16: single-run screening statistics are too noisy to inspect one at a time.

Alternatives.

- **Aggregate stage shift over all cancers** (screened and unscreened pooled). Simplest to present and the least informative, because it is exactly the comparison that length bias distorts.
- **The paired within-person stage shift as a first-class output** (its mean, median and full distribution, plus the detection-to-diagnosis interval). This is the recommendation: it is the quantity that actually drives the benefit, it is stable because it is paired, and it cannot be misread as a population-level shift.
- **The distribution of the screen lead time** rather than its mean. Shows the tail of very long sojourns that dominates the aggregate stage mix.
- **TNM-based shift** (tumour size, nodal involvement, distant metastasis) rather than the four-stage summary. The clinically meaningful description, and it would let the model’s stage shift be compared with trial results; it needs size and nodal data.
- **Age- and stage-standardised incidence and mortality by detection route** (the classical stage-shift analysis used in the screening literature). The standard that would make the model’s output comparable with published screening evaluations; it requires standardisation and a like-for-like denominator, both of which are straightforward additions.
- **“Proportion detected at stage I–II” as the headline screening metric.** Intuitive, easy to communicate, and it exposes the length-bias caveat immediately — which is either a feature or a trap depending on how it is worded.

12 Costs

Mechanism. The cost model is three lines of arithmetic. The screening cost is the number of tests performed times a unit test price. The treatment cost is the number of *cured* patients at each stage times that stage’s unit treatment price, summed over stages (1, 2, 3, 4 by stage in the packaged configuration). The total is their sum. In the reference run: 13 111 tests, screening cost 13 111, treatment cost 398, total 13 509.

Why this design. The prices are dimensionless because no trustworthy cost data are shipped, and a screening model whose headline message is a strategy *ranking* needs the relative prices to be defensible rather than the absolute ones. Encoding “a test costs one unit, treatment costs one to four units depending on stage” captures only the ordinal, uncontroversial clinical fact — later-stage treatment is more expensive — and leaves the user free to substitute real tariffs. For the same reason nothing in the calibration touches costs: they cannot silently absorb a misfit elsewhere.

Keeping the module to three lines is itself a design decision. A richer cost model would have to make assumptions (discount rate, perspective, follow-up protocols, complications) that the packaged data cannot support, and those assumptions would then drive the cost

comparison — whereas here the arithmetic can be checked by hand and the sensitivity of the ranking to the prices is transparent.

Why this design. *Three limitations that should be read as caveats on any cost figure.*

(a) Treatment is charged only for cured patients, so under a payer’s perspective the true cost of a screening programme is understated (patients who are treated and die are not charged) and the cost-per-life-saved arithmetic is not a cost-effectiveness ratio in the usual sense. (b) Three of the four price parameters are inert: `screening_price` (a per-person programme cost), `complications_price` and `reoccurrence_cost` are declared, stored, exposed in the interface and never used by the simulation. Treating a parameter set as configuration while some of it is decorative is a trap for a user who changes a price and sees nothing happen. (c) There is no discounting, no time preference, no perspective statement and no cost for a false positive, even though a false positive is the single most common costly event in a real programme.

Alternatives.

- **A full cost-effectiveness analysis** (cost per life-year or per QALY gained, discounting, incremental ratios against a comparator, willingness-to-pay thresholds, and probabilistic sensitivity analysis over the prices). This is what a health-economic consumer expects, and the model already computes the two ingredients (costs and years of life saved). It requires a perspective, a discount rate, utility weights and a comparator strategy — all of which are external decisions rather than data.
- **Treatment-pathway costing** (diagnosis work-up, surgery, chemotherapy, radiotherapy, palliative and terminal care, each with a state-specific cost). The standard in decision-analytic models, and it makes treatment costs respond to the stage at diagnosis in a clinically meaningful way; it needs pathway and tariff data.
- **A health-system perspective with real tariffs**, versus **a societal perspective** adding productivity losses, informal care and travel time. The two can differ by a factor, and the choice belongs to the user; the model should simply state which one it assumes (at present, neither).
- **Micro-costing of the screening episode** (invitation, kit, laboratory handling, and the follow-up procedure for a positive test). The dominant real-world cost driver at the margin, and the natural place for the currently inert `screening_price` and for the unmodelled false-positive consequences.
- **Charging for everyone treated, not only for those cured.** A one-line change that would make the cost figure behave as a payer expects, and would make the screening programme look correspondingly less attractive.
- **Cost by year since diagnosis** (the first year of treatment costs far more than subsequent years). A well-established pattern in the literature, and it would matter for any comparison that shifts diagnoses earlier in time.

13 Elements That Exist But Do Nothing

Mechanism. The configuration surface and the interfaces expose a number of elements that have no effect on any result. They are listed here rather than quietly omitted, because a user who changes a parameter and sees nothing happen has been misled by the model,

not by their own mistake. Every entry below was verified by searching for actual reads in the simulation path.

Element	What it was for	Actual status
Recurrence probability	recurrence after cure	Draws an exported flag only (<code>reoccurrence_probability = 0.2</code>); no effect on survival, death timing or cost.
Post-treatment mortality	extra mortality after treatment	Declared and never read (<code>treatment_mortality_adjustment = 1.0</code>).
Treatment complications	complication rate	Declared and never read (<code>complications = [0.1]</code>).
Complication cost	cost per complication	Declared and never read (<code>complications_price = 1.0</code>).
Recurrence cost	cost per recurrence	Declared and never read (<code>reoccurrence_cost = 1.0</code>).
Programme cost per person	per-person screening cost	Declared and never read (<code>screening_price = 1.0</code>); only the test price is charged.
False-positive rate	the test's false-positive rate	Read, stored and displayed but never applied (<code>test_fp</code> , <code>test_fp_selected</code>): the simulated rate is $0.1 \times TP$.
Risk factors	relative risks for exposures	Declared and never read by the simulation (<code>factors_rr</code> , <code>age_irrelevant_factors_rr</code> , <code>factor_age_correction</code>).
Factor accessors	helpers for those tables	Never called (<code>has_factor</code> , <code>get_factor_rr</code>).
Legacy configuration	the old key–value format	Reachable only by pointing the model at a <code>.txt</code> file (<code>from_legacy_txt</code> , <code>_csharp_to_python_name</code>); no packaged configuration uses it.
Duplicate cure helper	the age-cure step function	Never called (<code>get_age_cure_efficiency</code>); the simulation uses its own copy of the same logic.
Risk-to-distribution helper	annual risks $\rightarrow$ death-age distribution	Never called (<code>risks_to_distribution</code>); the model uses observed deaths directly.
Curve-evaluation helpers	evaluating the fitted Gompertz curve	Never called (<code>compute_output</code> , <code>compute_output_train</code> , <code>compute_train_ages</code> , <code>gompertz_v</code> , <code>compute_stage_age</code>); the fit and the simulation both use the closed-form crossing times.
Vectorised hazard draw	a vectorised first-success draw	Never called (<code>PiecewiseHazard.T_batch</code>); the population generator does this inline.
Scalar sampling helpers	single-value sampling	Never called (<code>Distribution.generate_single</code> , <code>RandomState.next_double</code> , <code>next_doubles</code>).

Why this design. Two principles justify listing this rather than deleting it silently. First, *an unimplemented mechanism should be visible as an unimplemented mechanism*. Several of these entries (treatment mortality, complications, recurrence) correspond to real clinical effects that the model does *not* represent; leaving them in the configuration while they do nothing invites a reader to assume they are active, and the resulting model is more dangerous than a smaller one that admits its scope. Second, *the size of the inert surface is a measure of the gap between intent and implementation* — here it is a fairly concentrated set of treatment-pathway and recurrence features, which is a useful signal about where the model's ambitions outran its inputs.

Why this design. *The weakness this exposes.* Inert parameters are a usability trap. A user who sets a false-positive rate, a complication rate or a screening price and observes no change has no way to tell a bug from a non-feature, and the interface’s willingness to display the parameter implies it works. The current code does carry in-line comments labelling several of them as inactive, which is the right instinct, but a comment is not a contract.

Alternatives.

- **Delete them.** Cleanest: a parameter set that contains only live parameters cannot mislead. The cost is losing the documented intent, which is real information for whoever implements the treatment pathway later.
- **Keep them, but label them in the interface as inert** (grey out, or annotate “not used in the current model”, with the reason). Preserves the intent and removes the trap — this is the smallest change that fixes the actual problem.
- **Fail loudly when an inert parameter is set to a non-default value** (a validation warning at load time). Guarantees that nobody believes a setting they just made is in force.
- **Implement the treatment pathway**, turning recurrence, complications and treatment mortality into active mechanisms. The correct long-run answer, and the only one that makes the parameters meaningful; it needs treatment and complication data.
- **Move the legacy loader out of the main path** (or delete it) so that a single configuration format remains. A second parsing path is a second surface for silent divergence — the two loaders would have to agree on the meaning of every field, and nothing tests that they do.

14 Calibration

14.1 Three separate likelihoods, and everything else by hand

Mechanism. When calibration is requested, the model solves three independent fitting problems.

1. **Diagnosis hazard** (6 coefficients): a Poisson likelihood on the observed age-specific case counts, $C(a) \sim \text{Poisson}(h(a)P(a))$, where $h(a)$ is the piecewise log-hazard of §4. Minimised with L-BFGS-B under ± 5 bounds; it converges in a fraction of a second.
2. **Gompertz shape and rate** (3 parameters K, C, B_{pop}): least squares on the three crossing times *derived from the stage mix* (§6.3), plus penalties that keep the carrying capacity above the terminal stage, keep the rate positive, and mildly regularise the capacity so the fit cannot run into a degenerate huge-capacity/tiny-rate corner. In the reference fit the residual is of order 10^{-3} — the targets are reproduced essentially exactly.
3. **Terminal-phase hazard** (1 parameter λ): a Poisson likelihood on the observed age-specific cancer deaths, $D(a) \sim \text{Poisson}(e^{\lambda}C(a))$ with $C(a) = I(a)P(a)$. The reference fit gives $\lambda = -1.05$, i.e. a rate of 0.35 per year.

Everything else is a configured value rather than an estimate: the lead-time mean, B_{std} , the stage treatment efficiencies, the age cure constants, the aggressiveness odds ratio, the screening schedule, participation, test accuracy and the prices. The stage-mix inversion that produces the crossing times is arithmetic, not optimisation, and it runs before the Gompertz fit.

Why this design. *(i) Splitting the problem is what keeps assumptions assumptions and estimates estimates.* A single joint fit over all parameters would be non-identifiable along the two flat ridges of §5.2 and §8.2: the optimiser could trade lead time against growth rate, or cure probability against terminal hazard, and improve the aggregate fit while moving the parameters arbitrarily far. Splitting the fit into three well-posed pieces means each parameter is pinned by a target that can actually pin it — and the quantities that no target can pin are *not* fitted at all: they are configured and then varied in the sensitivity analysis. That distinction, enforced by the structure of the code rather than by discipline, is the guardrail that matters most.

(ii) Each objective is the right likelihood for its data. Counts of cases and deaths are Poisson counts, not binomial proportions, and the discrete hazard of §4 is the natural link. Using the correct likelihood is not pedantry: the previous formulation used a binomial likelihood on proportions and — far worse — calibrated mortality against the *diagnosis* hazard rather than the cancer-death hazard, which made the mortality calibration meaningless while still looking plausible.

(iii) Small and smooth objectives are fast, reproducible and auditable. Six parameters and a smooth likelihood converge deterministically from a fixed starting point; one parameter is a line search. Nothing is fitted by simulation, so calibration takes milliseconds, results are exactly reproducible, and each likelihood value can be printed and checked.

(iv) Structural constraints are expressed as smooth penalties. Every threshold crossing must be reachable ($\exp K > s$) and the rate must be positive; encoding these as large quadratic penalties keeps the optimiser out of the degenerate region without a constrained solver or a rejection loop.

Why this design. *Three methodological weaknesses to state.* *(a)* The mortality likelihood is an *approximation*: it treats every case as subject to a constant death hazard, ignoring that only uncured patients die and that death follows diagnosis with a delay. The fitted λ therefore absorbs whatever the assumed cure probabilities get wrong — which is exactly the confound of §8.2, seen from the calibration side. *(b)* There is no train/validation split: the model is calibrated and judged on the same data. This is the weakest methodological point in the whole design, and the remedy is procedural rather than computational. *(c)* Not everything that *could* be fitted is fitted. The stage-mix calibration is dataset-specific by construction (the crossing times are derived from that dataset’s stage mix), so the only components transferable between datasets are the diagnosis hazard and the mortality hazard. A genuine external validation would therefore have to hold out one of those two, or hold out another population and accept that the staging calibration is re-derived — which limits what “validation” can mean here, and should be said wherever a validation claim is made.

Alternatives.

- **Joint Bayesian calibration with priors and posterior predictive checks** (MCMC or ABC). The statistically correct way to handle the flat ridges: the assumptions enter as priors and their consequences arrive as credible intervals instead of as a set of re-runs. Costs hours instead of milliseconds, and the headline numbers then depend on priors that are themselves not derived from data.
- **Simulated maximum likelihood** (computing the likelihood from simulation rather than analytically). Removes the mortality approximation and would make the cure-versus-terminal-hazard ridge explicit rather than hidden; the objective becomes noisy and far more expensive, which in turn requires many replications or a stochastic optimiser.
- **Approximate Bayesian computation or history matching**. Likelihood-free and well suited to a simulator whose analytic likelihood is only approximate; needs thousands of runs and an explicit tolerance, so it is a research workflow rather than an interactive one.
- **A surrogate model plus global optimisation** (the differential-evolution and Optuna paths already present in the code). Useful when the objective is multi-modal or noisy; it adds a hyper-parameter — the surrogate’s own fit — to the problem.
- **Manual / expert (Delphi) calibration**, as practised for several large cancer models. Transparent, and it can incorporate judgement the data do not carry; it is not reproducible and it is hard to audit.
- **Iterated sequential calibration to convergence** (fit the hazards, hold them, fit the staging, then revisit the hazards). Makes the coupling between components explicit; it needs a convergence criterion, and it can drift if the components are only weakly coupled.
- **A genuine hold-out protocol**: calibrate on one population or period and validate on another — the packaged USA and Germany datasets make this a one-line experiment for the transferable components — split temporally where multiple periods exist, and report the validation error as the headline quality metric. This is the highest-value methodological addition available without new data.

15 Randomness and Reproducibility

15.1 Two layers of randomness

Mechanism. The model uses two layers. *Population-level* draws (initial ages, natural death ages, the per-agent stream seeds) come from a single PCG64 generator seeded once per run and stored with the results. *Agent-level* draws inside the cancer-history kernel do not use that generator at all: each agent has a 64-bit stream identifier, and each quantity is drawn as $\text{uniform}(\text{seed}_i + k)$ for successive small offsets k , using a SplitMix64-style avalanche hash. Gaussian draws are Box–Muller on two uniforms from the same stream; Bernoulli draws are uniform comparisons. The generator state is *thread-local*, so concurrent runs in one process cannot re-seed each other, and the seed is logged with every run.

Why this design. (i) *Per-agent streams decouple agents from each other and from the order of work.* An agent’s draws depend only on its own identifier, so the results are

invariant to how the loop is laid out, to vectorisation, and to any future parallelism. That is what allows the kernel to iterate agents in whatever order is convenient and to be compiled as a tight loop.

(ii) *Neighbouring indices must decorrelate, because the kernel reads them consecutively.* Each agent draws many quantities from nearby offsets, so the map from index to value has to look like independent noise for offsets of 1, 2, 3... It did not. The previous linear congruential generator followed by a single xorshift step gave $\text{corr}(u(s), u(s+1)) \approx +0.36$ and $P(u(s+1) < u(s)) \approx 0.12$ — instead of 0 and 0.5 — which silently coupled the stage draw to the lead time, the cure draw, the cancer-death age and the aggressiveness label. Aggressive tumours, for example, were drawn almost only at stage IV. This was a real modelling error, not an aesthetic one: it changed the joint distribution of the model’s own variables. The avalanche hash fixes it, and a guard test now pins the invariant ($|\text{corr}| < 0.015$ and $|P - \frac{1}{2}| < 0.015$ at ten lags).

(iii) *Thread-local state is a correctness requirement for a web service.* The application serves concurrent runs from separate threads; a process-wide singleton let one run re-seed the stream another run was drawing from, which broke reproducibility only intermittently and therefore looked like flakiness rather than a bug.

(iv) *PCG64 for the population-level draws* is a modern, small-state, well-tested generator and it is the current NumPy default — the safe choice for quantities that are drawn once per agent in bulk.

(v) *Logging the seed makes every result reproducible in one command*, which is a prerequisite for the calibration, debugging and validation workflows, and for reproducing a figure quoted in a report.

Why this design. *Two honest qualifications.* First, the agent-level streams are *pseudo*-independent by construction: they are not formally independent, and their adequacy rests on the avalanche property being verified (which the guard test does) rather than on a theorem. Second, exactly reproducing a run requires both layers to be seeded consistently, so the seed reported with a result must be understood to control the agent-level seeds too — it does, because they are themselves drawn from the seeded population-level generator, but the dependency is worth stating.

Alternatives.

- **A single sequential RNG stream for everything.** Simplest possible design; it makes every result depend on the exact order of operations, which forecloses vectorisation and parallelism and means any code rearrangement changes all outputs. It is also what makes accidental correlations across variables easy to create and hard to notice.
- **A full generator object per agent** (each with its own PCG64 state). Statistically excellent and trivially independent; constructing one object per agent costs orders of magnitude more than hashing two 64-bit words, and the kernel is executed for every agent in the population.
- **Counter-based generators (Philox, Threefry).** Built precisely for this pattern — a counter plus a key streams out values — with documented statistical quality and cheap parallel jumps; the natural upgrade path, and it would remove the reliance on a hash’s avalanche behaviour.

- **Common random numbers across scenarios** (sharing the random shocks between the screened and unscreened arms to shrink the variance of the difference still further). The paired design of §10 already captures most of this benefit by storing the counterfactual rather than re-simulating it.
- **Replicate runs with reported intervals** rather than a single run (seed averaging). The honest way to present a stochastic model’s output, especially for tail quantities such as lives saved, where a single run’s count can move by several units; it multiplies the runtime by the number of replicates, which the interactive interface cannot currently afford.
- **Antithetic or quasi-random (Sobol) sampling** for the population-level draws. Reduces variance for integration-like quantities; much less useful for rare tail events, which is what the interesting outputs are.
- **A recorded draw-by-draw random-number log** for bit-level reproducibility under any future code change. Overkill today, but it is the standard a formally certified model is eventually held to.

16 Sensitivity Analysis as the Uncertainty Statement

Mechanism. The sensitivity analysis re-runs the whole model for multiples of the lead-time mean — 0.5, 0.75, 1.0, 1.25 and 1.5 by default — and reports the incidence, mortality, screening-mortality, survival and years-saved curves plus the stage distributions for every factor side by side. For a factor f the lead-time mean becomes $f\mu$ and the stage-crossing times are rescaled by the same factor, after which the Gompertz is refitted to the rescaled targets, so the model still reproduces the same stage mix at the new time scale. All other parameters stay at their fitted reference values, and the random seed is the same for every factor. Display smoothing is applied to the curves, never to the stored numbers.

Why this design. (i) *The honest response to an unidentifiable parameter is a range, not a value.* The lead time cannot be estimated from the available data (§5.2), so the only defensible thing to do with it is to show what the answer looks like across its plausible range. The sensitivity analysis is the model’s uncertainty statement about its one structural assumption — not an optional extra.

(ii) *Rescaling the crossing times together with the lead time is what keeps the comparison meaningful.* Because $t_s = -\mu \ln P(\text{stage} \geq s)$, changing μ while holding the crossing times fixed would change the stage mix and thereby confound the time-scale question with a staging change. Moving both together holds the observable stage mix constant, so the curves differ *because* of the time-scale assumption and nothing else.

(iii) *A common seed across factors is a variance-reduction device, and here it is essential.* Sharing the random shocks means the between-factor differences are the model’s genuine response to the assumption, with the Monte-Carlo noise largely cancelled out of the comparison. Without it, and with as few screening events as the reference run produces, the spread between curves could be dominated by sampling variation — a user would be reading noise as sensitivity.

(iv) *Smoothing is presentation-only* so that the numbers reported and exported remain exact, and the parallel execution path exists so that five full re-runs stay interactive, with an automatic threshold that avoids paying process start-up for jobs too small to benefit.

Why this design. *Two limitations that a reader of these plots should know.* First, *only the lead time varies*. The other hand-set quantities — B_{std} , the cure probabilities, the aggressiveness odds ratio, participation, test accuracy — are equally unverified, and a set of curves that varies only one axis can easily be read as “this is the only soft assumption”. Second, the analysis reports *curves* rather than a decision metric, so the quantity a decision-maker would actually want ranged (lives saved, or cost per life saved, under each factor) is not tabulated and has to be assembled from the metrics block by hand.

Alternatives.

- **A standardised sweep over every soft parameter**, with the results tabulated as decision metrics rather than as families of curves. One-at-a-time analysis is a lower bound on uncertainty (it cannot show interactions); it is easy to run because the model is fast, and it is the most valuable single addition available.
- **Global sensitivity analysis** (variance decomposition, Sobol indices, elementary-effects screening). Apportions the output variance across the assumptions and exposes interactions; needs a designed experiment over hundreds or thousands of runs, which is within reach of the current model but not of the current interface.
- **Probabilistic sensitivity analysis** (distributions on the soft parameters, Monte Carlo over draws, and intervals on the outcomes). The health-economic standard, and it converts “here are five curves” into “here is the distribution of the answer”; it requires defensible distributions, which for the hand-set parameters means priors.
- **Scenario analysis** (“optimistic test”, “pessimistic participation”, “slow tumours”) instead of a numeric sweep. More communicable to non-modellers and it forces the assumptions to be named; it is a presentation of the same information and loses the systematic coverage.
- **Value-of-information analysis** (which assumption, if researched, would most reduce decision uncertainty). The natural next question after a sensitivity analysis, and the one that turns the exercise into a research agenda; it needs a decision rule and a loss function.
- **Structural sensitivity**: swapping the mechanism itself (a size-threshold detection model instead of a lead-time model, per-stage dwell times instead of crossing times, an onset-driven natural history instead of a diagnosis-driven one) and comparing the outputs. No parameter sweep can reach this class of uncertainty, and it is where the largest unquantified risk lies.

17 Assumptions, Direction of Bias, and Falsifiability

17.1 The assumption register

Mechanism. Every simplification in the model is listed below with the direction in which it pushes the results. Where the direction is stated, it is the direction relative to a “true” model of the same question; where it is not determinable from the design alone, that is said instead of guessed.

Assumption	Direction of the bias if it is wrong
No pre-cancer stage (no adenoma–carcinoma sequence; every cancer arises invasive).	Overstates the screening benefit: removing a pre-cancerous lesion is an additional protective mechanism the model lacks.
No non-progressive lesions.	Overstates the benefit: every screen-detected cancer is assumed to be one that would have surfaced and could have killed, so overdiagnosis cannot be represented.
Diagnosis-driven natural history (no undiagnosed cancer deaths).	Understates the population’s cancer burden; harmless for the screening contrast, misleading for a burden estimate.
The absolute preclinical time scale (lead-time mean) is assumed, not estimated.	Not determinable — this is precisely why it is swept (§16). It is the model’s largest single assumption.
Growth heterogeneity (B_{std}) is set by hand.	Not determinable; too little makes the stage shift too clean and too favourable, too much does the reverse. Currently not swept.
Cure probabilities are set by hand and are confounded with the fitted terminal hazard.	The aggregate cancer mortality is matched either way; what is not identified is the composition — how many patients the “cure” actually rescues.
Aggressiveness does not affect growth.	Understates the biology; the observed stage gradient in survival is then partly a model construction rather than a mechanism.
One primary cancer per person; no second primaries or metachronous tumours.	Slightly understates incidence and mortality.
No treatment-related mortality, no complications.	Understates the harm side of both treatment and screening (the follow-up after a positive test).
No behavioural effects (a false positive does not change attendance; screening does not change lifestyle).	Not determinable; real programmes show both effects, in opposite directions.
Uniform treatment at a given stage, regardless of comorbidity, access or era.	Understates variation and inequality; the average may be biased in either direction depending on the population.
Closed population: no births, no migration.	Only matters over horizons long enough for the age structure to drift, which 30 years largely is not.
Annual discrete time; within-year ordering ignored.	Small, but systematic for the fastest terminal phases (mean terminal survival is only about three years).
Cancer death does not remove the agent from the simulated population; cause-specific rates use the general-population denominator.	Treats the two causes as independent; small at these rates, but it understates the correction that a competing-risk denominator would apply.
Age is the only exposure dimension: no sex, no region, no time trend, no risk factors.	The model represents one population in one period; applying it to trends, to another sex, or to a risk-stratified programme would be invalid rather than merely imprecise.
Benefit is measured within the simulation horizon.	Understates lifetime benefit, and makes the headline figure depend on the horizon.
Test performance is a constant (true-, false-positive) pair.	Omits size dependence and all false-positive harm, so the net benefit is slightly overstated.
Screen-detected and clinically detected cancers of the same stage are treated identically.	Ambiguous: screen-detected tumours often have better biology (understating the benefit), while the override of the clinical cure path can work the other way (§9.3).
No discounting and no time preference.	Overstates long-run benefits relative to costs when compared with a discounted analysis.
Risk factors, complications, recurrence and treatment mortality exist as parameters but do nothing (§13).	No effect; the risk is purely that a reader assumes they are active.

17.2 What would show the model to be wrong

Mechanism. Three classes of check are available, and they differ enormously in strength.

(i) *Internal consistency checks* — quantities the model is supposed to reproduce from data it was given: the age-specific incidence curve (fitted), the age-specific cancer mortality curve (fitted, but with the cure-versus-terminal-hazard confound of §8.2), and the stage mix at diagnosis. The third is the informative one, because only its three crossing-time *targets* are taken from the data: the stage mix itself is then produced by the interaction of the growth model and the lead time. In the reference run it is reproduced to within a few thousandths (0.225/0.250/0.284/0.241 against 0.220/0.256/0.308/0.218). A failure here would point squarely at the growth model, the crossing-time inversion, or the per-agent draws.

(ii) *External checks* — the ones that can actually falsify the model: a screening trial of the same test in the same age range, where the model’s benefit must agree in both sign and magnitude; the observed mix and rate of interval cancers in a real programme; the within-person stage shift measured in a screened cohort (§11.3); and the sojourn time inferred from screen-detected versus interval cancers, which would identify the one quantity the model has to assume. None of these are possible with the packaged aggregate data, and all of them are possible with trial or registry screening data.

(iii) *Structural checks* — consistency of the mechanism with what is known about the disease. The model has no pre-cancer stage and no non-progressive lesions, so if it predicts a benefit where a trial found none, those two omissions are the first suspects, in that order. A structural check can also be run as an experiment rather than as an argument: implement the omitted mechanism and see whether the conclusion survives.

Why this design. Falsifiability cannot be bolted on at the end: it has to be specified before the numbers are seen, or it degenerates into post-hoc rationalisation. The model is unusually well placed for this because it is fast and deterministic given a seed — a criterion can be checked in seconds — yet it currently has no pre-registered success criteria and no formal validation against anything outside its own calibration data. That is the gap between a working model and a defensible one.

Alternatives.

- **Pre-registered success criteria and analysis plan** (which outputs, which tolerance, which dataset, decided in advance). Cheap, purely procedural, and it converts validation from a narrative into a test.
- **A permanent benchmark suite** (recorded reference outputs under fixed seeds, checked on every change). The model already does this for the random generator; extending the same idea to the whole simulation would catch unintended changes — the ones that produce plausible-looking but different numbers, which are the dangerous ones.
- **Periodic recalibration against newly published data**, with the previous version’s predictions compared against the new observations and the discrepancy reported. Turns an annual data release into a published back-test.
- **An independent implementation of one component** (for example an independent recomputation of the crossing-time inversion and the Gompertz fit) to catch a

shared misconception between the specification and the code.

- **Decision-analytic validation** (does the model change a decision that would otherwise have been made?). The most relevant test for a screening model, and the only one that directly addresses whether the modelling error matters.

18 Summary: Every Element, Its Choice, Its Alternatives

The table below is the whole document in one place. For each modelling element it states the design that was chosen, the alternatives that were considered, and what would change upstream if the design were switched.

Element	Chosen design	Main alternatives	What changes upstream if switched
Model class	Agent-based microsimulation; both arms from the same agents	Cohort Markov, age-structured PDE, renewal equation, semi-Markov, compartmental ODE	The paired lives-saved accounting (the model's statistical efficiency) is lost; heterogeneity must be discretised into states.
Time	Annual discrete steps on an integer age grid	Continuous time with exponential clocks, monthly steps, event-driven simulation	Screening instants and terminal-phase timing become exact; the calibration targets stay annual, so the extra resolution is not controlled by data.
Population frame	Closed cross-sectional cohort, 30-year horizon	Birth cohort from age 0, open population with demographic projection, steady state	A birth cohort needs a century-long burn-in; an open population contaminates the screening contrast with demographic drift.
Initial ages	Observed age structure used as a probability mass	Uniform or parametric age profile, age bands, census microdata	Parametric profiles impose a shape; a uniform start destroys the dominant driver of incidence.
Non-cancer mortality	One sampled natural death age per agent, from observed deaths	Annual Bernoulli survival draws, parametric mortality law, cause-deleted life tables, frailty	Cause-deleted tables would separate the two causes properly (needs a cause-specific death column); frailty needs an unidentifiable distribution.
Onset hazard	Piecewise-constant log-hazard on age; the first Bernoulli success defines the diagnosis age	Smooth parametric hazard, splines, observed rates used directly, mechanistic carcinogenesis model	Splines are the natural refinement; a mechanistic model would make risk factors modelable but is not identifiable from incidence alone.
What counts as cancer	Diagnosed before natural death; agents past 98 excluded	Onset-driven natural history, a non-progressive lesion fraction, no exclusion at all	Onset-driven natural history would make overdiagnosis representable, and is the largest structural upgrade available.

Element	Chosen design	Main alternatives	What changes upstream if switched
Lead time	Exponential with an assumed mean (3.5 years)	Gamma/Weibull sojourn, deterministic lead time, per-stage dwell times, size-threshold detectability, estimation from trial data	Only trial or registry screening data identifies the scale; size-threshold detectability would unify detection with the growth curve.
Time-scale identification	Fix the scale; derive crossing times from the stage mix; sweep the scale	Bayesian prior and posterior, profile likelihood, fix the growth rate instead, two-parameter sojourn	A posterior converts one number plus a caveat into intervals on every output, at the cost of hours and of a prior.
Growth curve	Reduced Gompertz, $V(t) = \exp(K - \exp(C - Bt))$	Exponential, logistic, von Bertalanffy, Weibull/power law, mechanistic ODE, branching process, no growth model	A different curve changes the closed-form crossing times, and with them the entire staging calibration and the identifiability argument.
Growth heterogeneity	Log-normal B_{ind} with hand-set $B_{\text{std}} = 0.1$	No heterogeneity, Gamma growth rate, discrete growth classes, random effects on covariates	$B_{\text{std}} = 0$ removes length bias and makes the stage a deterministic function of the lead time.
Staging	Derived: the number of size thresholds crossed before diagnosis	Draw from the observed mix, ordinal regression on covariates, latent multi-state progression, TNM	Drawing the mix makes calibration a tautology; TNM would need size and nodal data.
Aggressiveness	Per-stage Bernoulli label modifying cure odds (OR 0.667)	Growth modifier, continuous latent score, mixture cure model, subtype-specific survival, no label	Making it a growth modifier double-counts against B_{std} and breaks the exact stage-mix fit.
Cure	Stage efficiency $\times$ age step function, capped, Bernoulli; cure means the cancer no longer decides death	Relative-survival tables used directly, mixture cure model estimated from follow-up, excess-hazard modelling, treatment pathways	Registry survival tables would remove the mechanism and the ability to ask why a patient survives.
Screening schedule	Age window plus fixed period; independent per-round participation	Continuous-time screening, correlated/sticky participation, never-attenders, programme scale-up	Sticky participation would make the benefit depend on reaching under-screened groups; the present design cannot express it.
Test performance	A (TP, FP) menu of which only TP is applied (FP = $0.1 \times \text{TP}$)	Sensitivity/specificity framing, size-dependent sensitivity, explicit follow-up costs	Making FP live is a bug fix; size-dependent sensitivity is what makes the length bias quantitative.
Screen detection	Re-uses the clinical cure model with the stage at detection; overrides the clinical path	Additive effect, route-specific cure parameter, route-specific survival from a trial	The override creates a harm channel; an additive design would remove it but understate the benefit.
False positives	Counted only, with no consequences	Follow-up procedures, costs, anxiety, effects on attendance	Modelling them is the correct direction, and it needs health-economic and behavioural data.

Element	Chosen design	Main alternatives	What changes upstream if switched
Death attribution	Derived from the stored ages; five exhaustive categories	Two-population comparison, same-seed twin run, relative survival, probability of causation	A two-population comparison loses the pairing and needs far more agents for the same precision.
Rates and outcomes	Counts by year and age; crude rates over person-years	Age standardisation, cumulative risk, interval bands, smoothing	Standardisation is necessary for cross-dataset comparison, and it is currently left to the reader.
Survival	Cause-specific from diagnosis, stratified by stage and aggressiveness, cured patients censored	All-cause survival, relative/net survival, excess hazard, parametric models, conditional survival	Relative survival is the registry standard and the only like-for-like comparison with published figures.
Stage shift	Both distributions reported; the within-person shift is the mechanism	Aggregate shift only, a paired shift as a first-class output, TNM shift, standardised stage-shift analysis	Reporting the paired shift would prevent the length-bias misreading that the marginals invite.
Costs	Tests $\times$ unit price plus cured patients $\times$ stage price; dimensionless	Full cost-effectiveness analysis, treatment-pathway costing, real tariffs, societal perspective, micro-costing	Charging everyone treated and adding follow-up costs would change the economics qualitatively, not merely in level.
Inactive elements	Documented rather than removed (§13)	Delete them, label them as inert in the interface, warn on non-default values, implement the treatment pathway	Implementing recurrence, complications and treatment mortality makes those parameters meaningful, and adds a real harm channel.
Randomness	PCG64 at population level, SplitMix64 hash per agent, thread-local state	A single sequential stream, a generator per agent, counter-based generators, common random numbers, replicates with intervals	Replicates with reported intervals are the honest presentation for tail quantities such as lives saved.
Calibration	Three separate likelihoods; everything unidentifiable is configured by hand	Joint Bayesian calibration, simulated likelihood, ABC/history matching, surrogate plus global optimisation, Delphi	A joint Bayesian fit would quantify the ridges, but would make the headline numbers prior-dependent.
Validation	None formal; the stage mix is the only informative internal check	Pre-registered criteria, hold-out population, benchmark suite, periodic back-testing, decision-analytic validation	Pre-registration is procedural and free, and it is the largest gap between this model and a defensible one.
Sensitivity	One axis (the lead-time scale), common seed, curves reported	A sweep over all soft parameters, global variance decomposition, probabilistic analysis, scenarios, value of information, structural swaps	Sweeping all soft parameters would show that the lead time is not the only unverified assumption.

19 Where to Read More

This document is the “why” of the model. Its two siblings cover the other angles:

- `model_for_dummies.pdf` — the same model in plain language, for a reader who wants to know what the software does rather than why it was built this way.
- `python_port_analysis.pdf` — the technical and statistical review: state variables, the statistical corrections applied during the port, performance, and the methodological improvements that would be needed for a formal policy-grade model.

The parameter file (`config/parameters.toml`) carries a comment against every hand-set parameter saying what it is and whether it is estimated; the sections above are the long form of those comments. The two structural assumptions to keep in view while reading any result are the assumed preclinical time scale and the assumed cure probabilities, because those are the two quantities that the data cannot pin down — and they are also the two that the sensitivity analysis, in its present one-axis form, does not fully cover.